Chapter 4: Small Oscillations

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In this chapter, we will study how physical systems behave in the vicinity of their stable equilibrium points. Such behavior is often referred to as *small oscillations* because the system exhibits oscillatory motion about its equilibrium position, with a displacement that remains small.

We will first examine the simplest case of oscillatory motion, in which a restoring force acts on the system, pushing it back toward its equilibrium position. We will then consider the effects of a retarding force, which dissipates mechanical energy and causes the amplitude of oscillations to decrease over time. Finally, we will study driven oscillations, in which an external periodic force acts on the system, and investigate how the resulting motion depends on the frequency of the driving force, including the phenomenon of resonance. For simplicity, we will restrict our attention to one-dimensional systems throughout this chapter.

1 Simple Oscillations

We begin by considering the motion of a particle under a conservative force described by a potential energy function $U(x)$. We make the following three assumptions:

(i) $U(x)$ is sufficiently smooth to allow a Taylor expansion about the equilibrium point.

(ii) The potential has a local minimum at $x = x_0$, with $U'(x_0) = 0$ and $U''(x_0) > 0$, so that x_0 is a stable equilibrium point.

(iii) The displacement from equilibrium, $|x - x_0|$, remains "sufficiently small".

Assumption (i) allows us to expand the potential energy about the equilibrium point x_0 using a Taylor series:

$$U(x) = U(x_0) + U'(x_0)(x - x_0) + \frac{1}{2}U''(x_0)(x - x_0)^2 + \frac{1}{6}U'''(x_0)(x - x_0)^3 + \dots \qquad (1)$$

According to assumption (ii), the first derivative vanishes at equilibrium, $U'(x_0) = 0$, while the second derivative is positive, $U''(x_0) > 0$. Furthermore, assumption (iii) allows us to

neglect the third and higher-order terms compared to the quadratic term. This is what we actually mean when we say that the displacement is "sufficiently small" in assumption (iii). Since the reference level of potential energy can be chosen arbitrarily, we may also set $U(x_0) = 0$ without loss of generality. We therefore obtain the quadratic approximation

$$U(x) \approx \frac{1}{2}k(x - x_0)^2, \tag{2}$$

where $k \equiv U''(x_0) > 0$. It is worth emphasizing that the approximation for the potential energy follows directly from our assumptions and is not limited to specific systems such as springs or pendulums. A wide range of physical systems can be described by a quadratic potential near their stable equilibrium points, provided that the second derivative of the potential at equilibrium is positive. This allows us to study the small oscillations of different physical systems using the same mathematical framework.

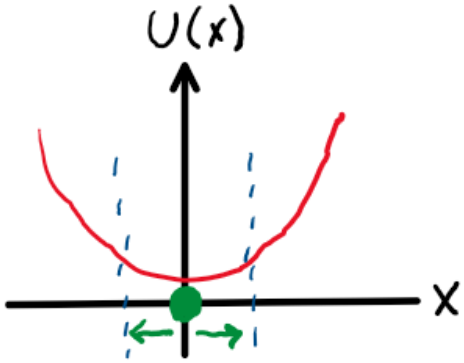


Figure 1: Small oscillations of a particle about a stable equilibrium point under potential energy $U(x)$. The particle oscillates about the equilibrium position x_0 with a small amplitude.

Our initial assumption of having a potential energy function directly means that we are considering a conservative force. Therefore, using the quadratic approximation, we can derive the force acting on the particle:

$$F(x) = -\frac{dU}{dx} \approx -k(x - x_0). \tag{3}$$

This is known as *Hooke's law*. The force is proportional to the displacement from equilibrium and always points toward the equilibrium position x_0. For this reason, it is called a *restoring force*. The constant k is called the *force constant* or *spring constant*, and it characterizes the stiffness of the system. A larger value of k corresponds to a stiffer system, which requires more force to achieve the same displacement from equilibrium.

From the experimental point of view, the force constant can be determined by measuring the force required to displace the particle from its equilibrium position and calculating the slope of the force-displacement curve as shown in Fig. 2. The force constant is a fundamental property of the system and depends on its physical characteristics, such as the material properties and geometry.

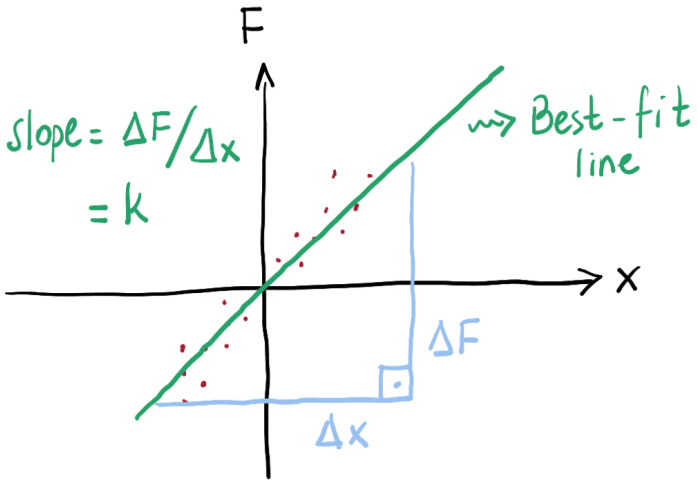


Figure 2: Empirical determination of the force constant k. The slope of the force-displacement curve gives the value of k. Note that the linear best-fit line is a model; the idea is there is always a sufficiently small region around the equilibrium point where the force is linear in the displacement. The data points may scatter according to different models as the displacement increaases.

1.1 Equation of Motion for Simple Oscillations

The dynamics of the particle are governed by Newton's second law:

$$F_{\text{net}} = m\ddot{x}, \tag{4}$$

where m is the mass of the particle. Assuming that the restoring force is the only net force acting on the particle, and choosing the origin at the equilibrium position ($x_0 = 0$), we obtain

$$m\ddot{x} = -kx. \tag{5}$$

Rearranging this equation gives

$$\ddot{x} + \omega_0^2 x = 0, \tag{6}$$

where we have introduced the positive parameter

$$\omega_0 \coloneqq \sqrt{\frac{k}{m}}. \tag{7}$$

This is a second-order linear homogeneous differential equation describing *simple harmonic motion*. The physical meaning of ω_0 will become clear when we solve this equation and examine the resulting motion.

1.2 Solution to the Equation of Motion

We know that the particle oscillates back and forth about the equilibrium position in a periodic way. We saw in Chapter 1 that the exponential function $e^{\lambda t}$ exhibits a wide range

of different types of behavior depending on how the parameter λ is chosen as summarized in the following table:

λ	Behavior of $e^{\lambda t}$
Real and positive ($\lambda > 0$)	Exponential growth
Real and negative ($\lambda < 0$)	Exponential decay
Purely imaginary ($\lambda = i$)	Sinusoidal oscillation
Complex, $\mathrm{Re}(\lambda) < 0$	Oscillation with exponentially decaying amplitude
Complex, $\mathrm{Re}(\lambda) > 0$	Oscillation with exponentially growing amplitude

We know that the solution to the differential equation in Eq. 6 must be oscillatory according to our assumptions. Therefore, it is natural[1] to look for a solution of the form $x(t) = e^{\lambda t}$, where λ will be determined by substituting this ansatz into the differential equation. Differentiating this ansatz twice and pluging it into the equation of motion as given by Eq. 6 gives

$$\lambda^2 e^{\lambda t} + \omega_0^2 e^{\lambda t} = 0 \implies \lambda^2 + \omega_0^2 = 0. \tag{8}$$

This equation is called the characteristic equation having the two purely imaginary roots

$$\lambda_{1,2} = \pm i\omega_0. \tag{9}$$

This leads to two different solutions $x_{1,2}(t) = e^{\pm i\omega_0 t}$. Not surprisingly, the both solutions have a pure complex factor of the time leading to oscillatory behavior.

A second-order linear homogeneous differential equation requires two initial conditions to determine a unique solution; therefore, the linear combination of two linearly independent solutions, containing two arbitrary constants, is sufficient to describe all possible solutions [2]. Thus, the general solution can be written as

$$x(t) = Ae^{i\omega_0 t} + Be^{-i\omega_0 t}, \tag{10}$$

where A and B are arbitrary constants determined by the initial conditions. In general, these constants can be complex. However, since $x(t)$ represents the position of a particle, it must be a real-valued function. To express the solution in a real form, we use Euler's formula,

$$e^{i\theta} = \cos\theta + i\sin\theta, \tag{11}$$

which also implies

$$e^{-i\theta} = \cos\theta - i\sin\theta. \tag{12}$$

Substituting these expressions into the general solution gives

$$\begin{aligned} x(t) &= A\left[\cos(\omega_0 t) + i\sin(\omega_0 t)\right] + B\left[\cos(\omega_0 t) - i\sin(\omega_0 t)\right] \\ &= (A+B)\cos(\omega_0 t) + i(A-B)\sin(\omega_0 t). \end{aligned} \tag{13}$$

[1] I admit that this is a handwaving argument. For the complete description, we need to study Laplace transforms which is beyond the scope of this text.

[2] The details about the construction of the general solution is a topic of linear algebra and again beyond the scope of this text.

At first glance, the coefficients $A + B$ and $i(A - B)$ may appear to be complex. However, we can show that they must be real by considering the initial conditions. Suppose that the particle has an initial position $x(0) = x_0$ and an initial velocity $\dot{x}(0) = v_0$. Evaluating the position at $t = 0$, we obtain

$$x(0) = A + B = x_0. \tag{14}$$

Differentiating the solution with respect to time gives

$$\dot{x}(t) = -\omega_0(A + B)\sin(\omega_0 t) + i\omega_0(A - B)\cos(\omega_0 t). \tag{15}$$

Evaluating the velocity at $t = 0$, we find

$$\dot{x}(0) = i\omega_0(A - B) = v_0, \tag{16}$$

and therefore

$$A + B = x_0, \qquad i(A - B) = \frac{v_0}{\omega_0}. \tag{17}$$

Since both the initial position x_0 and the initial velocity v_0 are real physical quantities, the combinations $A + B$ and $i(A - B)$ must also be real. Substituting these expressions back into the general solution, we finally obtain

$$x(t) = x_0 \cos(\omega_0 t) + \frac{v_0}{\omega_0}\sin(\omega_0 t). \tag{18}$$

This expression describes the motion of the particle for any given initial position x_0 and initial velocity v_0. For convenience, we can introduce two real constants, $C = x_0$ and $D = v_0/\omega_0$, and write the general solution in the more compact form

$$x(t) = C\cos(\omega_0 t) + D\sin(\omega_0 t). \tag{19}$$

The constants C and D are determined by the initial conditions.

1.3 Amplitude-Phase Form of the General Solution

This general solution given in Eq. 19 can also be expressed in terms of the *amplitude* and *phase* of the oscillation. To see this, we start from Eq. 19 and use Euler's formula to rewrite the trigonometric functions in terms of complex exponentials:

$$\begin{aligned} x(t) &= C\,\mathrm{Re}\left[e^{i\omega_0 t}\right] + D\,\mathrm{Re}\left[-ie^{i\omega_0 t}\right] \\ &= \mathrm{Re}\left[Ce^{i\omega_0 t} - iDe^{i\omega_0 t}\right] \\ &= \mathrm{Re}\left[(C - iD)e^{i\omega_0 t}\right], \end{aligned} \tag{20}$$

where Re denotes the real part of a complex number. We now introduce the complex number

$$Z := C - iD, \tag{21}$$

which is constant and called a *phasor* or *complex amplitude*. Recall that any nonzero complex number can be expressed in polar form as $Z = Re^{i\phi}$, where $R = |Z|$ is its magnitude and ϕ is its argument. Since $Z = C - iD$, its magnitude and argument can be determined from the real and imaginary parts:

$$R = \sqrt{C^2 + D^2}, \qquad \phi = \arctan\left(-\frac{D}{C}\right). \tag{22}$$

Substituting the polar form of Z into the expression for the displacement gives

$$\begin{aligned} x(t) &= \mathrm{Re}\left[Ze^{i\omega_0 t}\right] \\ &= \mathrm{Re}\left[Re^{i\phi}e^{i\omega_0 t}\right] \\ &= \mathrm{Re}\left[Re^{i(\omega_0 t+\phi)}\right]. \end{aligned} \tag{23}$$

Using Euler's formula, we obtain

$$x(t) = R\cos(\omega_0 t + \phi). \tag{24}$$

This is called the *amplitude-phase form* of the general solution. Recalling that $C = x_0$ and $D = v_0/\omega_0$, the amplitude and the initial phase can be expressed in terms of the initial conditions as

$$R = \sqrt{x_0^2 + \frac{v_0^2}{\omega_0^2}}, \qquad \phi = \arctan\left(-\frac{v_0}{\omega_0 x_0}\right). \tag{25}$$

The phasor representation also provides a geometric interpretation of the motion as shown in Fig. 3. The complex quantity $Re^{i(\omega_0 t+\phi)}$ represents a vector of constant length R rotating counterclockwise in the complex plane with angular velocity ω_0. The projection of this rotating vector onto the real axis gives the physical displacement $x(t)$ of the particle.

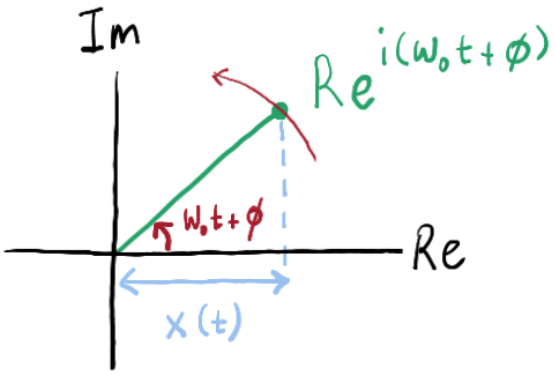


Figure 3: Phasor representation of simple harmonic motion. The complex vector $Re^{i(\omega_0 t+\phi)}$ rotates in the complex plane, and its projection onto the real axis gives the physical displacement $x(t)$.

1.4 Physical Interpretation of the General Solution

In the general solution given in Eq. (24), the constant $R \geq 0$ is called the *amplitude*, representing the maximum displacement of the particle from its equilibrium position. The constant ϕ is called the *initial phase* (or *phase constant*), which determines where in its oscillation cycle the particle is at $t = 0$. In other words, the initial phase determines at what point in the oscillation we start our chronometer. The constant ω_0 is the *angular frequency*, which determines how rapidly the particle oscillates. Recall from Eq. (7) that $\omega_0 = \sqrt{k/m}$, where k is the force constant and m is the mass of the particle. Thus, a stiffer system (larger k) or a lighter particle (smaller m) oscillates more rapidly. In Fig. 4, you can see how $x(t)$ varies with time.

The constant ω_0 is also called the *natural angular frequency* or *natural oscillation frequency* of the system, as it characterizes the intrinsic oscillatory behavior of the system in the absence of any external forces or damping. The natural angular frequency is a fundamental property of the system and depends only on its physical characteristics, such as the mass and stiffness. It is important to note that ω_0 is independent of the initial conditions, meaning that all particles with the same mass and force constant will oscillate with the same natural angular frequency, regardless of their initial position or velocity.

The motion is *periodic*, meaning that both the position and velocity of the particle repeat after a fixed time interval called the *period* T. For simple harmonic motion, the period is

$$T = \frac{2\pi}{\omega_0}. \tag{26}$$

The *frequency* f is the number of complete oscillations per unit time and is given by

$$f = \frac{1}{T} = \frac{\omega_0}{2\pi}. \tag{27}$$

The frequency is measured in hertz (Hz), while the angular frequency is measured in radians per second (rad/s).

As shown in Eq. (25), both the amplitude R and the initial phase ϕ are determined by the initial position and velocity of the particle. Unlike these two quantities, the angular frequency ω_0 depends only on the physical properties of the oscillator and not on its initial conditions. Consequently, the period of simple harmonic motion is independent of the amplitude of oscillation.

Finally, in the special case $R = 0$, the particle remains at its equilibrium position, $x(t) = 0$, and the phase ϕ is undefined.

1.5 Energy of Simple Oscillations

The mechanical energy of a particle undergoing simple oscillation

$$E = K + U = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2 \tag{28}$$

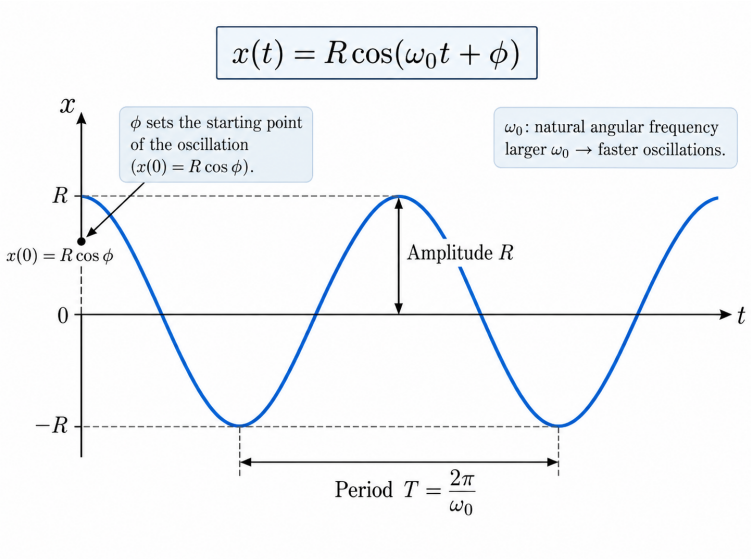


Figure 4: General solution of simple harmonic motion. The particle oscillates about the equilibrium position with amplitude R and angular frequency ω_0. The initial phase ϕ determines the starting point of the oscillation at $t = 0$.

Substituting the general solution $x(t) = R\cos(\omega_0 t + \phi)$ into the expressions for kinetic and potential energies, we find

$$K(t) = \frac{1}{2}m\left[-R\omega_0\sin(\omega_0 t + \phi)\right]^2 = \frac{1}{2}mR^2\omega_0^2\sin^2(\omega_0 t + \phi), \tag{29}$$

$$U(t) = \frac{1}{2}k\left[R\cos(\omega_0 t + \phi)\right]^2 = \frac{1}{2}kR^2\cos^2(\omega_0 t + \phi). \tag{30}$$

Using the relation $k = m\omega_0^2$, we can express the total energy as

$$E = \frac{1}{2}mR^2\omega_0^2\left[\sin^2(\omega_0 t + \phi) + \cos^2(\omega_0 t + \phi)\right] = \frac{1}{2}mR^2\omega_0^2. \tag{31}$$

By substituting $\omega_0^2 = k/m$, we can also express the total energy in terms of the force constant k and the amplitude R:

$$E = \frac{1}{2}kR^2. \tag{32}$$

Hence, the mechanical energy of a simple oscillator is constant and proportional to the square of the amplitude. This is a characteristic feature of conservative systems, where energy is conserved and oscillates between kinetic and potential forms without any loss as shown in the Fig. 5, with the maximum kinetic energy occurring when the particle passes through the equilibrium position ($x = 0$) and the maximum potential energy occurring at the turning points of the motion ($x = \pm R$).

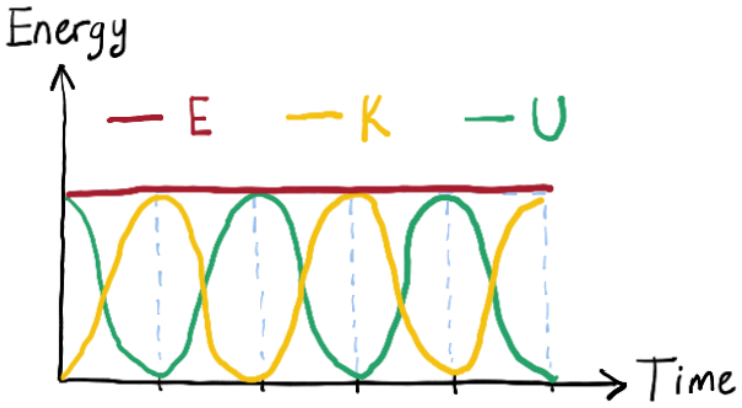


Figure 5: Oscillations of kinetic energy $K(t)$, potential energy $U(t)$, and total energy E for a particle undergoing simple harmonic motion. The total energy remains constant, while the kinetic and potential energies oscillate out of phase with each other.

2 Damped Oscillations

If the oscillating system is not isolated, it may experience a retarding force that dissipates mechanical energy. We will consider a model in which the retarding force is proportional to the velocity of the particle[3], which is a common situation in many physical systems. The retarding force can be expressed as

$$F_{\mathrm{r}} = -b\dot{x}, \tag{33}$$

where b is a positive constant called the *damping coefficient*. The negative sign indicates that the retarding force opposes the motion of the particle. The damping coefficient has units of force per unit velocity (N·s/m) and depends on the properties of the medium through which the particle moves, such as viscosity or air resistance. The total force acting on the particle is then the sum of the restoring force and the retarding force:

$$F_{\mathrm{net}} = F_{\mathrm{restoring}} + F_{\mathrm{retarding}} = -kx - b\dot{x}. \tag{34}$$

This leads to the equation of motion for damped oscillations:

$$m\ddot{x} + b\dot{x} + kx = 0. \tag{35}$$

Rearranging this equation gives

$$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = 0, \tag{36}$$

where we have introduced the positive parameters

$$\gamma \coloneqq \frac{b}{2m}, \qquad \omega_0 \coloneqq \sqrt{\frac{k}{m}}. \tag{37}$$

[3]Note that this is just a simple model that we can easily solve analytically. In real-life situations, the retarding force can be more complex and may depend on other factors such as the shape and size of the particle, the properties of the medium, and the frequency of the oscillation.

We would expect that the damping force would cause the amplitude of oscillations to decrease over time, eventually bringing the particle to rest at its equilibrium position. The parameter γ characterizes the strength of the damping, with larger values corresponding to stronger damping effects. The parameter ω_0 is the natural angular frequency of the undamped system, as we have seen in the previous section.

We know that the exponential function $e^{\lambda t}$ exhibits a wide range of different types of behavior depending on how the parameter λ is chosen. For the damped oscillation equation, we can look for a solution of the form $x(t) = e^{\lambda t}$, as we did it in Section 1.2 for simple oscillations. Plugging this into the differential equation in Eq. 36 leads to the characteristic equation

$$\lambda^2 + 2\gamma\lambda + \omega_0^2 = 0, \tag{38}$$

which has the two roots

$$\lambda_{1,2} = \frac{-2\gamma \pm \sqrt{(2\gamma)^2 - 4\omega_0^2}}{2} = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2}. \tag{39}$$

The roots can be real or complex, depending on the relative magnitudes of γ and ω_0, leading to three different cases: underdamped ($\gamma < \omega_0$), critically damped ($\gamma = \omega_0$), and overdamped ($\gamma > \omega_0$). Each case corresponds to a different type of motion, which we will analyze in detail in the following subsections.

2.1 Underdamped Oscillations ($\gamma < \omega_0$)

In the underdamped case, the damping is weak enough that the system still oscillates, but with a gradually decreasing amplitude. The roots of the characteristic equation are complex conjugates:

$$\lambda_{1,2} = -\gamma \pm i\omega_1, \tag{40}$$

where we have defined the *damped angular frequency*

$$\omega_1 \coloneqq \sqrt{\omega_0^2 - \gamma^2}. \tag{41}$$

The solution to the differential equation in this case is

$$x(t) = e^{-\gamma t}(Ae^{i\omega_1 t} + Be^{-i\omega_1 t}), \tag{42}$$

where A and B are constants determined by the initial conditions. According to Eq. 24, we can also express the solution in amplitude-phase form as

$$x(t) = (Re^{-\gamma t})\cos(\omega_1 t + \phi), \tag{43}$$

where R is the initial amplitude and ϕ is the initial phase. What we see inside the parantheses is the amplitude of the oscillation, which decays exponentially over time due to the damping effect. On the other hand, the cosine term describes the oscillatory motion with angular

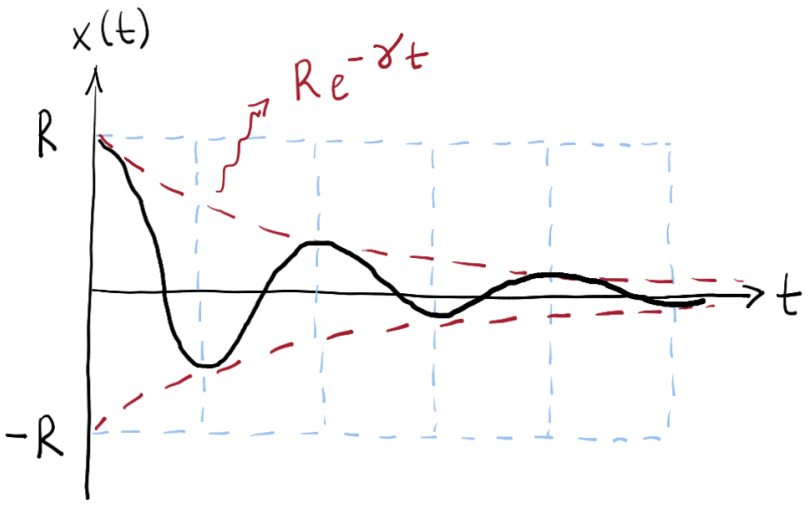


Figure 6: Underdamped oscillations of a particle. The amplitude of oscillation decays exponentially over time due to the damping effect, while the particle continues to oscillate with a damped angular frequency ω_1.

frequency ω_1. Therefore, the motion is a combination of exponential decay and sinusoidal oscillation as shown in Fig. 6.

Directly followed from the definition of ω_1, the damped angular frequency ω_1 is slightly less than the natural angular frequency ω_0, indicating that the oscillations occur at a lower frequency than in the undamped case

$$\omega_1 = \sqrt{\omega_0^2 - \gamma^2} < \omega_0.$$

Strictly speaking, the word "frequency" attributed to periodic motion. Therefore, calling ω_1 a frequency is not strictly correct, since the motion is not strictly periodic due to the damping. However, it is common practice to refer to ω_1 as the damped angular frequency, as it characterizes the rate of oscillation in the presence of damping for a given time interval.

We know that the mechanical energy of a simple oscillator is constant and proportional to the square of its amplitude. In the case of underdamped oscillations, the amplitude decreases exponentially over time, resulting in a corresponding loss of mechanical energy. Since both the displacement and velocity acquire the same exponential factor $e^{-\gamma T}$ after one complete period T, the mechanical energy satisfies

$$E(t+T) = E(t)e^{-2\gamma T}. \tag{44}$$

Consequently, after n complete periods,

$$E(nT) = E_0 e^{-2\gamma nT}, \tag{45}$$

where E_0 is the initial mechanical energy. The decrease in mechanical energy is due to dissipation, typically involving its conversion into heat or other forms of energy.

2.2 Critically Damped Oscillations ($\gamma = \omega_0$)

In the critically damped case, the damping is just strong enough to prevent oscillations. The roots of the characteristic equation are real and equal:

$$\lambda_{1,2} = -\gamma. \tag{46}$$

Therefore, we do not have two distinct solutions. We mentioned in Sec. 1.2 that the general solution for a second-order linear differential equation with constant coefficients is a linear combination of two independent solutions. In this case, we have only one independent solution, which is $e^{-\gamma t}$. To find a second independent solution, we return to the equation of motion, which, for $\gamma = \omega_0$, takes the form

$$\ddot{x} + 2\gamma\dot{x} + \gamma^2 x = 0. \tag{47}$$

Multiplying both sides by $e^{\gamma t}$, we obtain

$$e^{\gamma t}\ddot{x} + 2\gamma e^{\gamma t}\dot{x} + \gamma^2 e^{\gamma t}x = 0. \tag{48}$$

Note that the left-hand side is precisely the second derivative of the product $e^{\gamma t}x(t)$, as can be verified by applying the product rule twice:

$$\frac{d^2}{dt^2}\left[e^{\gamma t}x(t)\right] = e^{\gamma t}\ddot{x} + 2\gamma e^{\gamma t}\dot{x} + \gamma^2 e^{\gamma t}x. \tag{49}$$

Therefore, the equation of motion can be rewritten as

$$\frac{d^2}{dt^2}\left[e^{\gamma t}x(t)\right] = 0. \tag{50}$$

Integrating twice with respect to time gives

$$e^{\gamma t}x(t) = A + Bt, \tag{51}$$

where A and B are arbitrary constants. Solving for $x(t)$, we obtain

$$x(t) = (A + Bt)e^{-\gamma t}. \tag{52}$$

We can now identify the two linearly independent solutions:

$$x_1(t) = e^{-\gamma t}, \qquad x_2(t) = te^{-\gamma t}. \tag{53}$$

Indeed, neither solution can be expressed as a constant multiple of the other, so they are linearly independent. The constants A and B can be determined from the initial conditions $x(0) = x_0$ and $\dot{x}(0) = v_0$. Substituting these conditions gives

$$A = x_0, \qquad B = v_0 + \gamma x_0. \tag{54}$$

Thus, the motion of a critically damped oscillator is described by

$$x(t) = \left[x_0 + (v_0 + \gamma x_0)t\right] e^{-\gamma t}. \tag{55}$$

Since both $e^{-\gamma t}$ and $te^{-\gamma t}$ approach zero as $t \to \infty$, the particle eventually returns to its equilibrium position without undergoing oscillations. Depending on the initial conditions, the particle may cross the equilibrium position once, but it cannot oscillate back and forth around it.

2.3 Overdamped Motion ($\gamma > \omega_0$)

In the overdamped case, the damping is sufficiently strong to prevent oscillations. The roots of the characteristic equation are real and distinct:

$$\lambda_{1,2} = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2}. \tag{56}$$

Since $\gamma > \omega_0$, both roots are negative, with $\lambda_2 < \lambda_1 < 0$. The general solution to the differential equation is

$$x(t) = Ae^{\lambda_1 t} + Be^{\lambda_2 t}, \tag{57}$$

where A and B are constants determined by the initial conditions.

Since both exponential terms decay with time, the particle eventually approaches its equilibrium position. However, the two terms decay at different rates. Since $\lambda_1 > \lambda_2$, the term $e^{\lambda_1 t}$ decays more slowly and dominates the motion at sufficiently long times, provided $A \neq 0$.

Unlike the critically damped case, overdamping generally results in a slower approach to equilibrium. Although the particle may cross the equilibrium position once, depending on its initial conditions, it cannot undergo oscillatory motion about equilibrium.

Since $\lambda_2 < \lambda_1 < 0$, both exponential terms decay with time. At sufficiently long times, the term $e^{\lambda_1 t}$ dominates, provided $A \neq 0$. Notice that

$$\lambda_1 = -\frac{\omega_0^2}{\gamma + \sqrt{\gamma^2 - \omega_0^2}} > -\omega_0, \tag{58}$$

where we have used $\gamma > \omega_0$. Thus, the dominant exponential decays more slowly than the exponential factor $e^{-\omega_0 t}$ appearing in the critically damped solution. Overdamping therefore generally produces a slower relaxation toward equilibrium.

Furthermore, the particle can cross equilibrium only when $x(t) = 0$, which, for nonzero A and B, requires

$$e^{(\lambda_1 - \lambda_2)t} = -\frac{B}{A}. \tag{59}$$

Since the left-hand side is a strictly increasing function of time, this equation can have at most one solution. Consequently, the particle may cross equilibrium once, depending on its initial conditions, but it cannot undergo oscillatory motion.

2.4 Comparison of Damping Regimes

The underdamped motion is the only case that exhibits oscillatory behavior, while the critically damped and overdamped cases do not oscillate. The critically damped case represents the fastest approach to equilibrium without oscillation, while the overdamped case results in a slower approach to equilibrium. The underdamped case exhibits oscillatory motion with exponentially decaying amplitude, while the critically damped and overdamped cases do not

oscillate and approach equilibrium at different rates. The comparison of the three damping regimes is illustrated in Fig. 7.

The critically damped case is often considered the optimal damping scenario for systems that require a quick return to equilibrium without overshooting, such as in automotive suspension systems, certain types of mechanical clocks, pneumatic-actuated systems and screen door closure systems. The overdamped case, while slower, may be preferred in situations where overshooting must be avoided at all costs, even if it means a longer time to reach equilibrium.

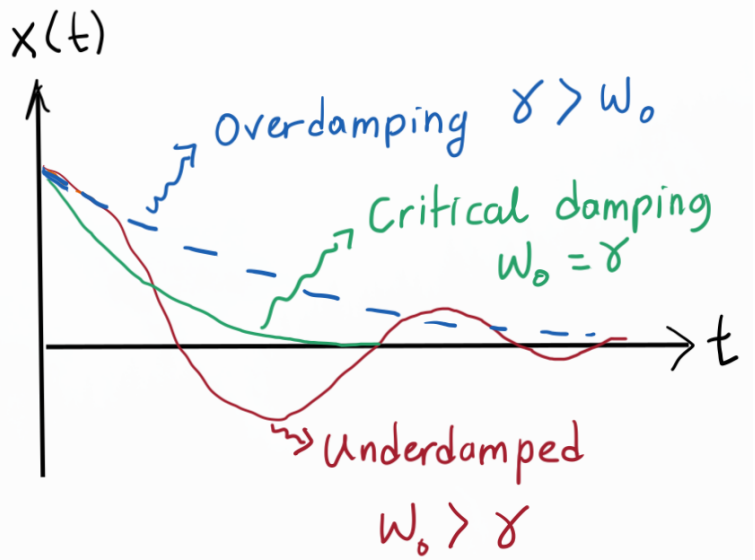


Figure 7: Comparison of the three damping regimes: underdamped, critically damped, and overdamped. The underdamped case exhibits oscillatory motion with exponentially decaying amplitude, while the critically damped and overdamped cases do not oscillate and approach equilibrium at different rates.

2.5 Quality Factor

The *quality factor* Q is a dimensionless quantity that characterizes how weakly an oscillating system is damped. We know that an undamped oscillator conserves its mechanical energy, whereas a damped oscillator gradually loses energy. A natural way to characterize the strength of damping is therefore to compare the mechanical energy stored in the oscillator with the energy dissipated during one complete oscillation.

For weak damping ($\gamma \ll \omega_0$), the quality factor can be expressed approximately as

$$Q \approx 2\pi \frac{E(t)}{\Delta E}, \tag{60}$$

where $E(t)$ is the mechanical energy at tan arbitrary time and ΔE is the energy dissipated over the subsequent oscillation period T. The factor 2π is included by convention. A high quality factor corresponds to weak damping, meaning that the oscillator loses only a small

fraction of its mechanical energy during each cycle. To relate the quality factor to the damping parameter γ, we recall that the displacement and velocity of an underdamped oscillator is given by

$$x(t) = R_0 e^{-\gamma t}\cos(\omega_1 t + \phi), \qquad \dot{x}(t) = -R_0 e^{-\gamma t}\left[\gamma\cos(\omega_1 t + \phi) + \omega_1 \sin(\omega_1 t + \phi)\right]. \tag{61}$$

After one complete oscillation period $T = 2\pi/\omega_1$, the trigonometric functions return to their original values. Consequently, both the displacement and velocity are multiplied by the same exponential factor:

$$x(t+T) = e^{-\gamma T}x(t), \qquad v(t+T) = e^{-\gamma T}v(t). \tag{62}$$

Since the total mechanical energy is given by $E(t) = \frac{1}{2}mv^2(t) + \frac{1}{2}kx^2(t)$, we obtain

$$\begin{aligned} E(t+T) &= \frac{1}{2}mv^2(t+T) + \frac{1}{2}kx^2(t+T) \\ &= e^{-2\gamma T}\left[\frac{1}{2}mv^2(t) + \frac{1}{2}kx^2(t)\right]. \end{aligned} \tag{63}$$

Therefore,

$$E(t+T) = E(t)e^{-2\gamma T}. \tag{64}$$

The mechanical energy dissipated during one complete oscillation is

$$\begin{aligned} \Delta E &= E(t) - E(t+T) \\ &= E(t)\left(1 - e^{-2\gamma T}\right). \end{aligned} \tag{65}$$

For weak damping ($\gamma \ll \omega_0$), we have $2\gamma T \ll 1$. Using the first-order Taylor approximation

$$e^{-2\gamma T} \approx 1 - 2\gamma T, \tag{66}$$

we obtain

$$\frac{\Delta E}{E(t)} \approx 2\gamma T. \tag{67}$$

Substituting this result into the expression for the quality factor gives

$$Q \approx \frac{2\pi}{2\gamma T} = \frac{\pi}{\gamma T} = \frac{\omega_1}{2\gamma}. \tag{68}$$

Since $\omega_1 = \sqrt{\omega_0^2 - \gamma^2} \approx \omega_0$ for weak damping, we obtain

$$Q \approx \frac{\omega_0}{2\gamma}. \tag{69}$$

For the linearly damped harmonic oscillator, this last expression is conventionally adopted as the exact definition of the quality factor:

$$Q \coloneqq \frac{\omega_0}{2\gamma}. \tag{70}$$

Its interpretation in terms of the energy dissipated per oscillation, however, is valid approximately in the weak-damping regime as described above. Rearranging the energy-based expression for the quality factor gives

$$\frac{\Delta E}{E(t)} \approx \frac{2\pi}{Q}. \tag{71}$$

This relation provides a direct physical interpretation of Q: the larger the quality factor, the smaller the fraction of mechanical energy dissipated during each oscillation. For example, an oscillator with $Q = 100$ loses approximately

$$\frac{\Delta E}{E(t)} \approx \frac{2\pi}{100} \approx 0.063, \tag{72}$$

or about 6% of its mechanical energy per cycle.The quality factor also allows us to express the damped angular frequency ω_1 in terms of the natural angular frequency ω_0. Using Eq. (70), we find

$$\omega_1 = \sqrt{\omega_0^2 - \gamma^2} = \omega_0 \sqrt{1 - \frac{1}{4Q^2}}. \tag{73}$$

For example, when $Q = 10$, we obtain $\omega_1 \approx 0.9987\omega_0$. Thus, for weak damping, the oscillation frequency remains very close to the natural frequency, even though the amplitude gradually decreases with time. Even for $Q = 2$, we have $\omega_1 \approx 0.968\omega_0$, which is still a relatively small deviation from the natural frequency. To obtain another physical interpretation of Q, let us determine how many oscillation cycles occur before the amplitude decreases to $1/e$ of its initial value. The amplitude decays according to

$$R(t) = R_0 e^{-\gamma t}, \tag{74}$$

where R_0 is the initial amplitude. The time required for the amplitude to decrease to R_0/e is therefore

$$t_{1/e} = \frac{1}{\gamma}. \tag{75}$$

The number of oscillation cycles during this time is

$$N = \frac{t_{1/e}}{T} = \frac{\omega_1}{2\pi\gamma}. \tag{76}$$

For weak damping, $\omega_1 \approx \omega_0$, and hence

$$N \approx \frac{Q}{\pi} \implies Q \approx \pi N. \tag{77}$$

This result shows that the quality factor is directly proportional to the number of oscillation cycles required for the amplitude to decrease to $1/e$ of its initial value. For example, an oscillator with $Q = 100$ completes approximately 32 cycles during this time. Thus, the quality factor provides two complementary physical interpretations of damping: it characterizes both the fraction of mechanical energy dissipated during each oscillation cycle and the number of cycles over which the oscillation amplitude significantly decreases.

3 Phase Space and Phase Trajectories

The *state* of a particle undergoing one-dimensional Newtonian motion is specified by its position $x(t)$ and velocity $\dot{x}(t)$, provided that the force is determined by these quantities and possibly time. These quantities can be regarded as coordinates in a two-dimensional space called *phase space*, with position along the horizontal axis and velocity along the vertical axis. The state of the particle is then represented by a vector in phase space,

$$\mathbf{X}(t) = \begin{bmatrix} x(t) \\ \dot{x}(t) \end{bmatrix}, \tag{78}$$

and as time progresses, the tip of this vector traces out a curve called a *phase trajectory*, as shown in Fig. 8(a). The phase trajectory provides a visual representation of the motion of the particle, showing how its position and velocity change together.

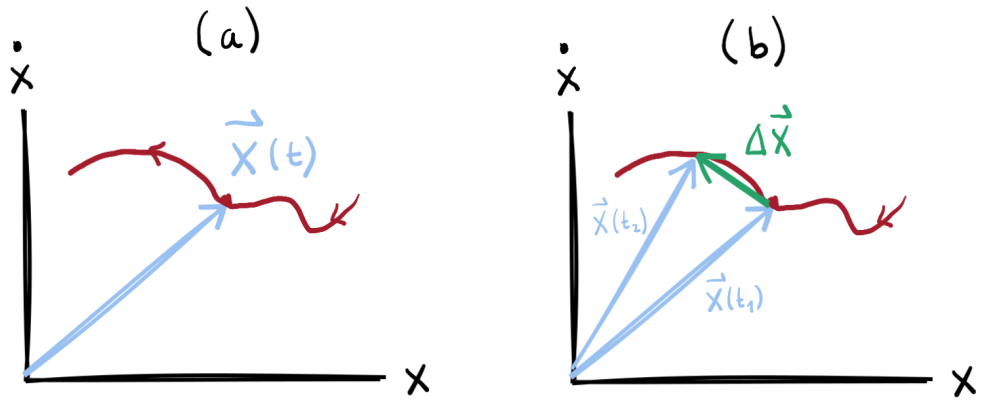


Figure 8: (a) State of a mechanical system in phase space. The state of the particle is represented by a vector $\mathbf{X}(t)$, whose tip traces out a phase trajectory as time progresses. (b) Displacement of the state vector in phase space defining its average "velocity," $\mathbf{V}_{\text{avg}} = \frac{\Delta \mathbf{X}}{\Delta t}$.

Note that this vector lives in the abstract phase space with coordinates $(x, \dot{x})$. A point in this space represents a state, not a position in physical space. Moreover, since position and velocity have different physical dimensions, the plotted length and direction of the state vector depend on the chosen axis scales. Nevertheless, we can take the time rate of change of this vector by taking the time rate of change of each component, as shown in Fig. 8(b). Thus, we get the velocity of the state in phase space,

$$\frac{d\mathbf{X}}{dt} = \begin{bmatrix} \dot{x}(t) \\ \ddot{x}(t) \end{bmatrix}, \tag{79}$$

where the first component is the velocity and the second component is the acceleration of the particle. The second component is determined by Newton's second law.

3.1 Time Evolution in Phase Space for Oscillatory Motion

For undamped, unforced simple harmonic motion, the position measured from equilibrium and the velocity are given by

$$x(t) = R\cos(\omega_0 t + \phi), \tag{80}$$
$$\dot{x}(t) = -R\omega_0 \sin(\omega_0 t + \phi). \tag{81}$$

Eliminating time from these equations leads to the equation of the phase trajectory. For $R > 0$, dividing the first equation by R and the second by $R\omega_0$, squaring both equations, adding them, and using the identity $\sin^2\theta + \cos^2\theta = 1$, we obtain

$$\frac{x^2}{R^2} + \frac{\dot{x}^2}{R^2\omega_0^2} = 1. \tag{82}$$

This is the equation of an ellipse in phase space, with semi-axes R and $R\omega_0$. Recall that the total mechanical energy of a simple harmonic oscillator is conserved and can be expressed in terms of its amplitude as

$$E = \frac{1}{2}kR^2 = \frac{1}{2}m\omega_0^2 R^2 \implies R^2 = \frac{2E}{k} = \frac{2E}{m\omega_0^2}. \tag{83}$$

Substituting this expression into the equation of the ellipse gives

$$\frac{x^2}{2E/k} + \frac{\dot{x}^2}{2E/m} = 1. \tag{84}$$

Thus, for fixed m and k, the semi-axes of the ellipse are determined by the total mechanical energy E of the oscillator, as shown in Fig. 9(a). The phase trajectories form a family of ellipses, each corresponding to a different positive value of the conserved mechanical energy E, as shown in Fig. 9(b). As the particle oscillates, its state moves clockwise along the corresponding ellipse, completing one revolution during each oscillation period. For $E = 0$, the particle remains at rest at equilibrium, and the phase trajectory reduces to the origin.

The time evolution of the state vector is governed by the equation

$$\frac{d\mathbf{X}}{dt} = \begin{bmatrix} \dot{x}(t) \\ \ddot{x}(t) \end{bmatrix} = \begin{bmatrix} \dot{x}(t) \\ -\omega_0^2 x(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\omega_0^2 & 0 \end{bmatrix} \begin{bmatrix} x(t) \\ \dot{x}(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\omega_0^2 & 0 \end{bmatrix} \mathbf{X}(t), \tag{85}$$

where the matrix

$$\hat{S} \equiv \begin{pmatrix} 0 & 1 \\ -\omega_0^2 & 0 \end{pmatrix} \tag{86}$$

is called the *system matrix*. By using this terminology, we can now write the time evolution of the system in a compact form as

$$\dot{\mathbf{X}} = \hat{S}\mathbf{X}. \tag{87}$$

This is a first-order linear differential equation for the state vector $\mathbf{X}(t)$, which can be solved using standard techniques that are beyond the scope of this text. However, even without

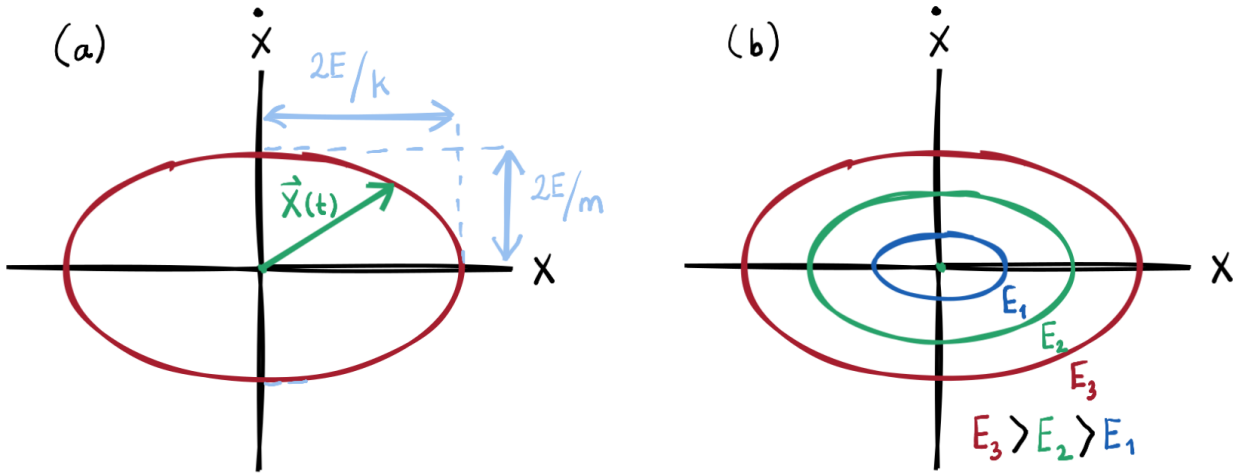


Figure 9: (a) Phase trajectory of a simple harmonic oscillator in phase space. (b) A family of elliptical trajectories, with semi-axes determined by the total mechanical energy E of the oscillator for fixed m and k. Note that phase trajectories belonging to different energies never intersect each other. If they did, a given state would have two different energies, which is not possible.

getting into the details of the solution of this equation, several conclusions can be drawn from this simple relation, as listed below:

Conclusion 1: The time rate of change of the state vector $\mathbf{X}$ can be obtained by combining its components according to the prescription given by the system matrix $\hat{S}$:

$$\frac{d}{dt}\begin{bmatrix} x(t) \\ \dot{x}(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\omega_0^2 & 0 \end{bmatrix}\begin{bmatrix} x(t) \\ \dot{x}(t) \end{bmatrix}. \tag{88}$$

Conclusion 2: Note that the system matrix is only a function of the natural angular frequency ω_0 of the system. Therefore, the evolution law is determined solely by the properties of the system, and not by the initial conditions. The initial conditions determine which phase trajectory the system will follow in phase space and the starting point on that trajectory. Thus, the actual time-dependent state depends on both the system properties and the initial conditions.

Conclusion 3: Once we are given the initial conditions at some arbitrary time t_0, $x(t_0)$ and $\dot{x}(t_0)$, or in other words, if we are given the initial "state" $\mathbf{X}(t_0)$ of the system, we can approximately determine the state of the system at a slightly later time $t_0 + \Delta t$:

$$\mathbf{X}(t_0 + \Delta t) \approx \mathbf{X}(t_0) + \dot{\mathbf{X}}(t_0)\Delta t = \mathbf{X}(t_0) + \hat{S}\mathbf{X}(t_0)\Delta t. \tag{89}$$

By repeating this process, we can approximately determine the state of the system at later times $t_n = t_0 + n\Delta t$:

$$\mathbf{X}_{n+1} = \mathbf{X}_n + \hat{S}\mathbf{X}_n\Delta t, \tag{90}$$

where $\mathbf{X}_n$ is the approximate state at time t_n. In this way, we can approximately trace out the phase trajectory of the system in phase space. This procedure is called the *forward Euler*

method. The quality of the approximation improves as Δt becomes smaller over a fixed time interval. However, this approximation does not conserve the oscillator's energy. Indeed, its component equations are

$$x_{n+1} = x_n + v_n \Delta t, \qquad v_{n+1} = v_n - \omega_0^2 x_n \Delta t, \tag{91}$$

where v_n is the approximate particle velocity at time t_n. Substitution into the energy expression gives

$$E_{n+1} = \frac{1}{2} m \left(v_{n+1}^2 + \omega_0^2 x_{n+1}^2 \right) = \left(1 + \omega_0^2 \Delta t^2 \right) E_n. \tag{92}$$

Thus, for a nonzero initial energy and any fixed nonzero time step, the approximate trajectory gradually spirals outward instead of remaining on an ellipse. Small time steps are therefore necessary for this method to approximate the motion over a finite time interval. There are more sophisticated numerical methods that conserve energy, but they are beyond the scope of this text.

Note that the exact future state of the system depends only on the current complete state and the system properties, and not separately on the history of the system. This is a deterministic example of the *Markov property*: once the present state is specified, knowledge of past states provides no additional information about the future. Here, this property holds because position and velocity together form a complete state governed by a closed first-order differential equation.

Conclusion 4: Note again that the system matrix $\hat{S}$ does not change in time once the system properties are set, and it acts on any state vector in phase space according to the same rule. It maps each state $\mathbf{X}$ to its time derivative $\hat{S}\mathbf{X}$. We can therefore think of phase space as consisting of infinitely many possible states represented by dots, and assign a "velocity vector" $\hat{S}\mathbf{X}$ to each of these dots. This defines a vector field in phase space, as shown in Fig. 10.

This velocity vector field does not change in time. Therefore, if we pictorially represent the state of the system by a "ball" in phase space, the matrix $\hat{S}$ defines a local rule at each point specifying how the ball moves through that point. The ball follows these local velocity vectors continuously and traces out a phase trajectory as time progresses. This is a useful way of visualizing the dynamics of the system under different initial conditions.

For the damped oscillation, time rate of change of the state vector becomes

$$\frac{d\mathbf{X}}{dt} = \begin{bmatrix} \dot{x}(t) \\ \ddot{x}(t) \end{bmatrix} = \begin{bmatrix} \dot{x}(t) \\ -2\gamma\dot{x} - \omega_0^2 x(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\omega_0^2 & -2\gamma \end{bmatrix} \mathbf{X}(t). \tag{93}$$

4 Driven Oscillations

We will consider now that the particle which does damped oscillatory motion is also subjected to an external periodic driving force, $F(t)$. These kind of oscillations are called *driven oscillations*.

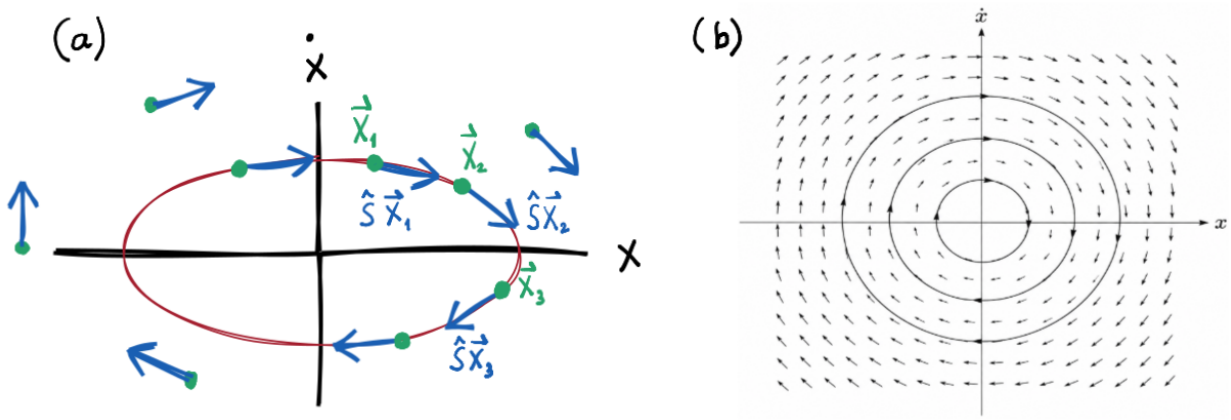


Figure 10: Phase-space vector field. (a) Starting from an initial state, the tip of the state vector traces out a phase trajectory as it moves through the vector field. (b) The system matrix $\hat{S}$ defines a velocity vector at each point in phase space, specifying the rates of change of position and velocity at that state.

An intuitive example of driven oscillations is a child on a swing. The child can be thought of as a damped oscillator, with the swing providing the restoring force. If the child pumps their legs at the right frequency, they can increase the amplitude of their swinging motion. In this case, the external driving force is provided by the child's legs, which are applying a periodic force to the swing.

The net force acting on the particle is the sum of the restoring force, the damping force, and the external driving force:

1. Restoring force: $F_{\text{restoring}} = -kx(t)$
2. Damping force: $F_{\text{damping}} = -b\dot{x}(t)$
3. External driving force: $F_{\text{driving}} = F(t) = F_0 \cos(\omega t)$

In general, the external driving force can be any function of time, but for simplicity, we will consider a sinusoidal driving force of the form $F(t) = F_0 \cos(\omega t)$, where F_0 is the amplitude of the driving force and ω is its angular frequency. The total force acting on the system and the corresponding equation of motion can be written as

$$F_{\text{net}} = F_{\text{r}} + F_{\text{d}} + F_{\text{drive}} = -kx(t) - b\dot{x}(t) + F_0 \cos(\omega t) = m\ddot{x}(t). \tag{94}$$

Rearranging this equation gives the equation of motion for a driven damped harmonic oscillator:

$$m\ddot{x}(t) + b\dot{x}(t) + kx(t) = F_0 \cos(\omega t). \tag{95}$$

Or, in terms of the natural angular frequency ω_0 and the damping constant γ, we can write

$$\ddot{x}(t) + 2\gamma\dot{x}(t) + \omega_0^2 x(t) = \frac{F_0}{m} \cos(\omega t). \tag{96}$$

This is a second-order linear nonhomogeneous differential equation with constant coefficients. It is nonhomogeneous because of the explicit time-dependent driving force on the right-hand side. Before getting into the solution of this equation, let us first discuss what would we expect from the solution.

Assume a weakly damped system was oscillating on its own approximately with frequency ω_0. If an external force starts "shaking" the system, in the beginning, the system will resist to obey the external force because of its intertia and previous motion. However, as the external force persistently continues to act on the system, the system will eventually "give in" and start to oscillate with the same frequency as the external force. This is called *steady-state motion*. In addition to this steady-state motion, there will also be a transient motion that depends on the initial conditions of the system. The transient motion will eventually die out due to damping, leaving only the steady-state motion.

From this point of view, we would expect two different behaviors from the general solution of the equation of motion: a transient motion that decays over time and a steady-state motion that persists indefinitely. The transient motion is called the *complementary solution* $x_{\mathrm{c}}(t)$ and it is the same as the solution of the homogeneous equation (the equation without the driving force). The steady-state motion is called the *particular solution* $x_{\mathrm{p}}(t)$ and it is a specific solution of the nonhomogeneous equation that depends on the form of the driving force. Therefore, the general solution of the equation of motion can be expressed as the sum of the complementary and particular solutions:

$$x(t) = x_{\mathrm{c}}(t) + x_{\mathrm{p}}(t). \tag{97}$$

We already know from previous section that the complementary solution dies out as $t \to \infty$:

$$\lim_{t\to\infty} x_{\mathrm{c}}(t) = 0 \implies \lim_{t\to\infty} x(t) = x_{\mathrm{p}}(t). \tag{98}$$

Two examples of this combined behavior of particular and complementary solutions is given in Fig 11. Following the above discussion, we actually know the form of the particular solution. Since the driving force is sinusoidal, we expect the particular solution to also be sinusoidal with the same frequency as the driving force, but with a different amplitude and phase. Therefore, we can write

$$x_{\mathrm{p}}(t) = A\cos(\omega t - \delta), \tag{99}$$

where A is the amplitude of the steady-state motion and δ is the phase difference between the driving force and the steady-state motion. This form of the solution provides us with the most general description of the steady-state motion of a driven damped harmonic oscillator regarding the points discussed above. The amplitude A and phase δ can be determined by substituting the particular solution into the equation of motion and solving for A and δ. This can be done by differentiating the particular solution twice and substituting it into the equation of motion. However, the calculations can be quite involved because of the trigonometric functions. Instead, we can so the same calculations in a more elegant way by using complex numbers.

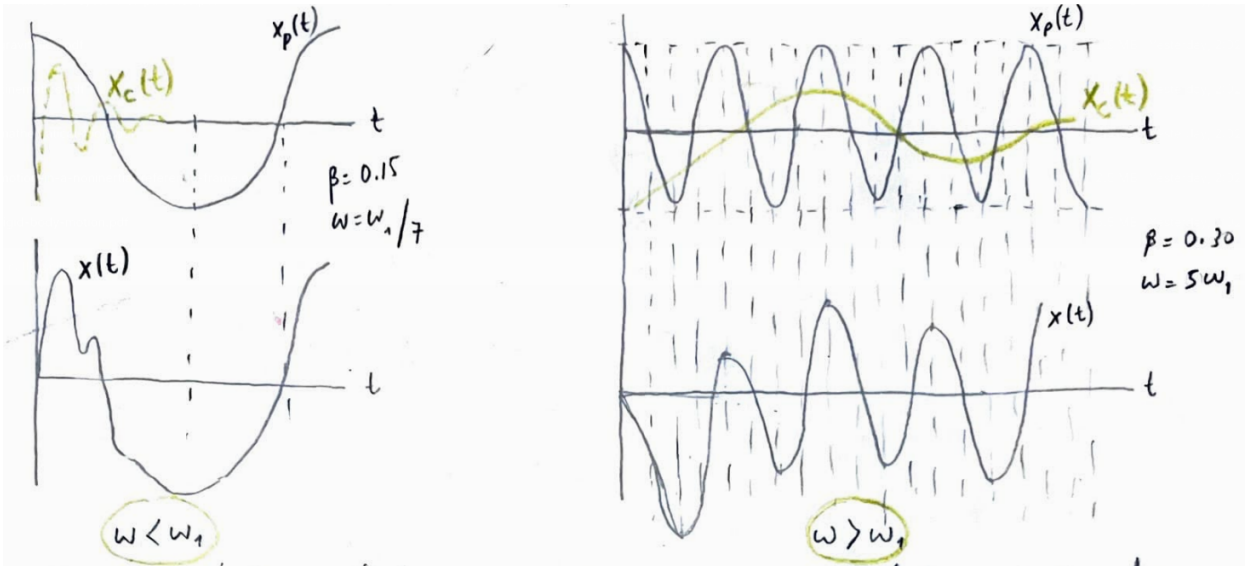


Figure 11: Two examples of driven damped harmonic oscillators with different driving frequencies. The transient motion dies out over time, leaving only the steady-state motion.

Instead of using Eq. (99) as a solution ansatz, let us switch to the complex plane by expanding it with a complex component:

$$z_{\mathrm{p}}(t) = A\cos(\omega t - \delta) + iA\sin(\omega t - \delta) = Ae^{i(\omega t - \delta)}. \tag{100}$$

From now on, we also think that the equation of motion is also extended to the complex plane by allowing the position $x(t)$ to be a complex function $z(t)$. Therefore, we can write the equation of motion (Eq. (96)) in the complex plane as

$$\ddot{z}(t) + 2\gamma\dot{z}(t) + \omega_0^2 z(t) = \frac{F_0}{m}e^{i\omega t}. \tag{101}$$

Note that we effectively changed nothing by switching to the complex plane since derivation is a linear operation, and actually, it corresponds to considering the same differential equation twice, once for the real part and once for the imaginary part,

$$\frac{d^2}{dt^2}\left[x_{\mathrm{p}}(t) + iy_{\mathrm{p}}(t)\right] + 2\gamma\frac{d}{dt}\left[x_{\mathrm{p}}(t) + iy_{\mathrm{p}}(t)\right] + \omega_0^2\left[x_{\mathrm{p}}(t) + iy_{\mathrm{p}}(t)\right] = \frac{F_0}{m}\left[\cos(\omega t) + i\sin(\omega t)\right]. \tag{102}$$

where $x_{\mathrm{p}}(t)$ and $y_{\mathrm{p}}(t)$ are the real and imaginary parts of the complex solution $z_{\mathrm{p}}(t)$, respectively. As you can see real and imaginary part do not mix together. The real part of the complex solution corresponds to the physical solution we are interested in, while the imaginary part is an auxiliary function that helps us simplify the calculations. The rest of the calculation is straightforward. We can differentiate the complex solution twice and substitute it into the equation of motion to solve for the amplitude A and phase δ:

$$\dot{z}_{\mathrm{p}}(t) = i\omega Ae^{i(\omega t - \delta)}, \tag{103}$$

$$\ddot{z}_{\mathrm{p}}(t) = -\omega^2 Ae^{i(\omega t - \delta)}. \tag{104}$$

Substituting these expressions into the equation of motion gives

$$\left[-\omega^2 + 2i\gamma\omega + \omega_0^2\right] A e^{i(\omega t - \delta)} = \frac{F_0}{m} e^{i\omega t}. \tag{105}$$

Dividing both sides by $e^{i\omega t}$ and rearranging gives

$$\left[-\omega^2 + 2i\gamma\omega + \omega_0^2\right] A = \frac{F_0}{m} e^{i\delta} = \frac{F_0}{m} \left[\cos(\delta) + i \sin(\delta)\right]. \tag{106}$$

Real and imaginary parts of this equation can be separated to give two equations for the amplitude A and phase δ:

$$A\left(\omega_0^2 - \omega^2\right) = \frac{F_0}{m} \cos(\delta), \tag{107}$$

$$2\gamma\omega A = \frac{F_0}{m} \sin(\delta). \tag{108}$$

Squaring both equations and adding them gives

$$A^2 \left[\left(\omega_0^2 - \omega^2\right)^2 + (2\gamma\omega)^2\right] = \left(\frac{F_0}{m}\right)^2. \tag{109}$$

Taking the square root of both sides gives the amplitude of the steady-state motion:

$$A = \frac{F_0/m}{\sqrt{\left(\omega_0^2 - \omega^2\right)^2 + (2\gamma\omega)^2}}. \tag{110}$$

The phase difference δ can be determined by taking the ratio of the imaginary and real parts of the equation:

$$\tan(\delta) = \frac{2\gamma\omega}{\omega_0^2 - \omega^2} \implies \delta = \arctan\left(\frac{2\gamma\omega}{\omega_0^2 - \omega^2}\right). \tag{111}$$

The amplitude and phase equations are the most important results of this section, and they provide a complete description of the steady-state motion of a driven damped harmonic oscillator. The amplitude equation shows how the amplitude of the steady-state motion depends on the driving frequency ω, the natural frequency ω_0, the damping constant γ, and the amplitude of the driving force F_0. The phase equation shows how the phase difference between the driving force and the steady-state motion depends on the same parameters. In Fig. 12, we plot the amplitude and phase equations as functions of the driving frequency ω.

4.1 How to read amplitude and phase equations

(a) F_0 is the amplitude of the driving force and it is in the numerator of the amplitude equation. Therefore, increasing the driving force will increase the amplitude of the steady-state motion or vice versa.

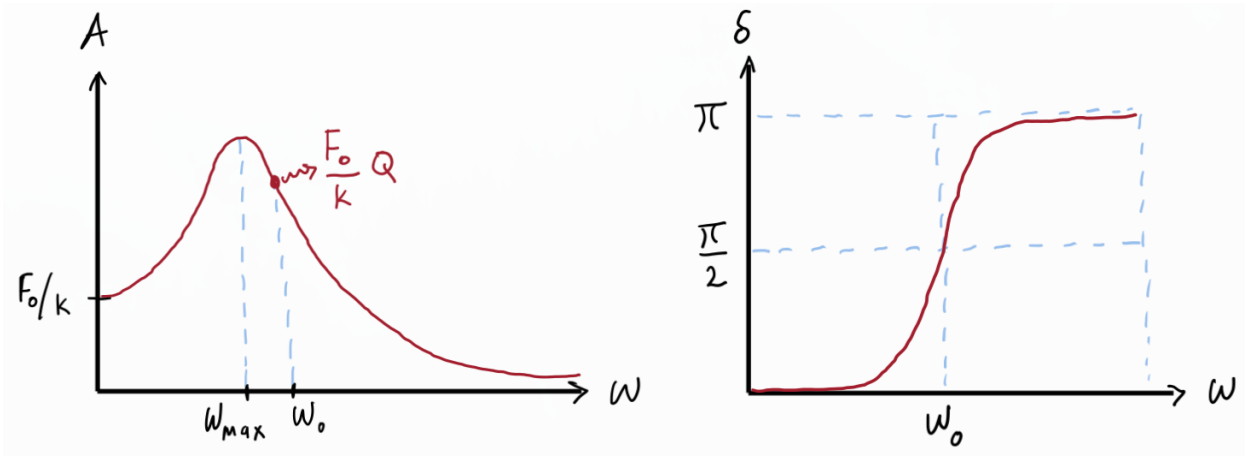


Figure 12: Amplitude and phase difference of the steady-state motion as a function of the driving frequency ω.

(b) γ is the damping constant and it is in the denominator of the amplitude equation. Therefore, increasing the damping will decrease the amplitude of the steady-state motion or vice versa.

(c) What if ω, the angular frequency of the driving force, is a very small number, $\omega \ll 1$? It physically means that driving force oscillates so slow that the particle directly follows the driving force. At each instant of time, we would expect the restoring force to be approximately equal to the driving force, i.e. $kx(t) \approx F_0 \cos(\omega t)$. Therefore, the amplitude of the steady-state motion should be approximately $A \approx F_0/k$. This is exactly what we get from the amplitude equation in this limit:

$$A \approx \frac{F_0/m}{\sqrt{\omega_0^2}} = \frac{F_0/m}{\omega_0^2} = \frac{F_0/m}{k/m} = \frac{F_0}{k}. \tag{112}$$

Also, since the particle is directly following the driving force, we would expect the phase difference to be approximately zero, $\delta \approx 0$. This is exactly what we get from the phase equation in this limit:

$$\delta \approx \arctan\left(\frac{0}{\omega_0^2}\right) = \arctan(0) = 0. \tag{113}$$

(d) What if ω, the angular frequency of the driving force, is equal to the natural angular frequency of the system, $\omega = \omega_0$?

$$A = \frac{F_0/m}{\sqrt{(2\gamma\omega_0)^2}} = \frac{F_0/m}{2\gamma\omega_0}. \tag{114}$$

Remember that the quality factor is defined as $Q = \omega_0/2\gamma$. Therefore, we can write the amplitude at resonance as

$$A_{\text{res}} = \frac{F_0/m}{2\gamma\omega_0} = \frac{F_0/m}{\omega_0^2}\frac{\omega_0}{2\gamma} = \frac{F_0/k}{2\gamma/\omega_0} = \frac{F_0/k}{1/Q} = Q\frac{F_0}{k}. \tag{115}$$

Therefore, the amplitude at resonance is Q times larger than the amplitude at low frequencies. This is a very important result, and it shows that the quality factor is a measure of how much the amplitude of the steady-state motion is amplified at resonance. The phase difference at resonance is

$$\delta = \arctan\left(\frac{2\gamma\omega_0}{\omega_0^2 - \omega_0^2}\right) = \arctan(\infty) = \frac{\pi}{2}. \tag{116}$$

(e) What if ω, the angular frequency of the driving force, is a very large number, $\omega \gg 1$? It physically means that driving force oscillates so fast that the particle cannot follow the driving force. At each instant of time, we would expect the particle to be approximately at rest, i.e. $x(t) \approx 0$. Therefore, the amplitude of the steady-state motion should be approximately $A \approx 0$. This is exactly what we get from the amplitude equation in this limit:

$$A \approx \frac{F_0/m}{\sqrt{\omega^4}} = \frac{F_0/m}{\omega^2} \approx 0. \tag{117}$$

In Eq. 111, as we increase ω, the parameter of the arctan function goes to zero from the left. Therefore, the phase difference approaches π from below, $\delta \to \pi^-$. This means that the particle is approximately at rest, and the driving force is approximately equal to the restoring force, but in the opposite direction, $F_0\cos(\omega t) \approx -kx(t)$.

(f) The maximum of the amplitude occurs not at the resonance frequency ω_0, but at a slightly lower frequency $\omega_{\max}$:

$$\frac{dA}{d\omega} = \frac{F_0}{m}\left[-\frac{1}{2}\left(4\gamma^2\omega^2 + (\omega^2 - \omega_0^2)^2\right)\right]^{-3/2}\left(8\gamma^2\omega + 4\omega(\omega^2 - \omega_0^2)\right) = 0. \tag{118}$$

$$8\gamma^2\omega + 4\omega(\omega^2 - \omega_0^2) = 0 \implies 2\gamma^2 + \omega^2 - \omega_0^2 = 0 \implies \omega_{\max} = \sqrt{\omega_0^2 - 2\gamma^2}. \tag{119}$$

Therefore, $\omega_{\max} < \omega_0$. We can also write $\omega_{\max}$ in terms of the quality factor Q:

$$\omega_{\max} = \sqrt{\omega_0^2 - 2\gamma^2} = \omega_0\sqrt{1 - \frac{1}{2Q^2}}. \tag{120}$$

If we plug $\omega_{\max}$ into the amplitude equation, we can find the maximum amplitude:

$$A_{\max} = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega_{\max}^2)^2 + (2\gamma\omega_{\max})^2}} = \frac{F_0/m}{\sqrt{(2\gamma^2)^2 + (2\gamma\sqrt{\omega_0^2 - 2\gamma^2})^2}} = \frac{F_0/m}{2\gamma\sqrt{\omega_0^2 - \gamma^2}}. \tag{121}$$

In terms of the quality factor Q, we can write

$$A_{\max} = \frac{F_0/m}{2\gamma\sqrt{\omega_0^2 - \gamma^2}} = \frac{F_0/m}{2\gamma\omega_0\sqrt{1 - \frac{1}{4Q^2}}} = \frac{F_0/k}{\sqrt{1 - \frac{1}{4Q^2}}}\frac{1}{2\gamma/\omega_0} = \frac{F_0/k}{\sqrt{1 - \frac{1}{4Q^2}}}\frac{1}{1/Q} = \frac{F_0}{k}\frac{Q}{\sqrt{1 - \frac{1}{4Q^2}}}. \tag{122}$$

For example, for a very low quality factor like $Q = 5$:

$$\omega_{\max} = \sqrt{\omega_0^2 - 2\gamma^2} = \omega_0\sqrt{1 - \frac{1}{2Q^2}} = \omega_0\sqrt{1 - \frac{1}{50}} \approx 0.99\omega_0. \tag{123}$$

Therefore, even for this very low quality factor, the maximum amplitude occurs at a frequency that is very close to the natural frequency of the system. For a higher quality factors the difference between ω_{max} and ω_0 becomes even smaller. When we compare the maximum amplitude A_{max} with the amplitude at resonance A_{res}, we can see that they are very close to each other for a low quality factor like $Q = 5$:

$$A_{\text{max}} = \frac{F_0}{k}\frac{Q}{\sqrt{1-\frac{1}{4Q^2}}} = \frac{F_0}{k}\frac{5}{\sqrt{1-\frac{1}{100}}} \approx 5.025\frac{F_0}{k}, \quad A_{\text{res}} = Q\frac{F_0}{k} = 5\frac{F_0}{k}. \tag{124}$$

In Fig. 13, we plot the amplitude response of a driven damped harmonic oscillator for different quality factors. As we can see, the maximum amplitude occurs at a frequency that is very close to the natural frequency of the system, and the difference between ω_{max} and ω_0 becomes smaller as the quality factor increases.

Even if the maximum amplitude occurs at a frequency ω_{max}, the amplitude at resonance A_{res} is still a very good approximation for the maximum amplitude A_{max}, especially for high quality factors. For this reason, by definiton, the natural oscillation frequency ω_0 is also called the *resonance frequency* of the system, and the amplitude at resonance A_{res} is also called the *resonance amplitude* of the system.

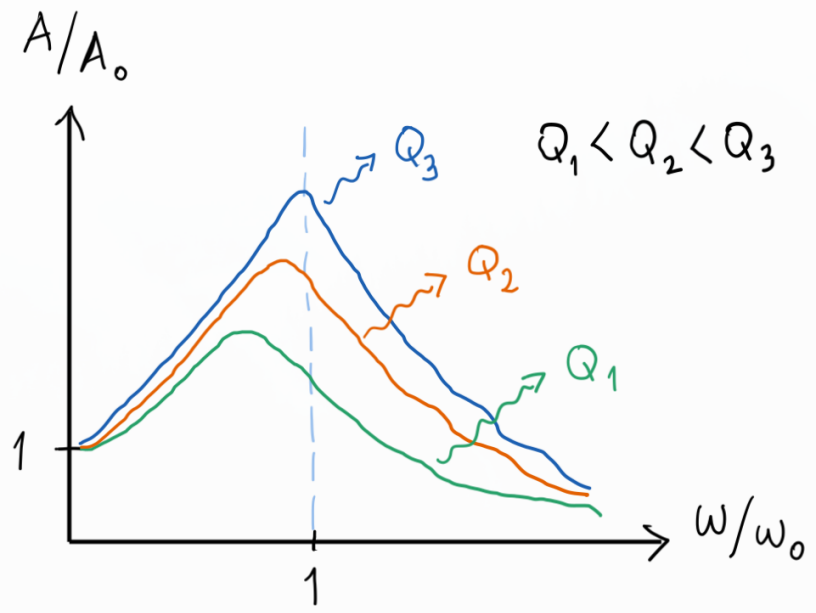


Figure 13: Amplitude response of a driven damped harmonic oscillator for different quality factors. The maximum amplitude occurs at a frequency that is very close to the natural frequency of the system, and the difference between ω_{max} and ω_0 becomes smaller as the quality factor increases.

5 Worked Examples

5.1 Work, Potential Energy, and Energy Conservation in a Spring

We are given a spring-mass system with a spring constant k and a mass m. By using the definition of work,

(a) Calculate the potential energy of the spring at a point x_0 with respect to its equilibrium position, where $U(0) = 0$.

(b) Calculate the work done by an external agent to bring the mass quasistatically from a point x_0 to the equilibrium position of the spring.

(c) If the force applied by an external agent is twice as large as the spring force in magnitude but opposite in direction at every position, calculate the work done by the external agent to bring the mass from a point x_1 to x_2, where $x_1 < x_2$.

Solution:

According to Hooke's law, the force exerted by the spring is

$$F_s(x) = -kx. \tag{125}$$

When the mass is moved quasistatically, the external force balances the spring force:

$$F_{\text{ext}}(x) = -F_s(x) = kx. \tag{126}$$

(a) The work done by an external agent in moving the mass quasistatically from the equilibrium position to x_0 is stored as potential energy in the spring, since the kinetic energy of the mass does not change. Therefore,

$$U(x_0) = W_{\text{ext}} = \int_0^{x_0} F_{\text{ext}}(x)\,dx = \int_0^{x_0} kx\,dx = \boxed{\frac{1}{2}kx_0^2}. \tag{127}$$

(b) When the mass is moved quasistatically from x_0 to the equilibrium position, the external force remains equal and opposite to the spring force. Thus,

$$W_{\text{ext}} = \int_{x_0}^{0} F_{\text{ext}}(x)\,dx = \int_{x_0}^{0} kx\,dx = \boxed{-\frac{1}{2}kx_0^2}. \tag{128}$$

The external agent does negative work because its force opposes the displacement toward equilibrium. The spring does an equal amount of positive work:

$$W_s = -W_{\text{ext}} = \frac{1}{2}kx_0^2. \tag{129}$$

Consequently, the decrease in potential energy is equal to the work done by the spring.

(c) The external force is twice as large as the spring force in magnitude and acts in the opposite direction:

$$F_{\text{ext}}(x) = -2F_s(x) = 2kx. \tag{130}$$

Therefore, the work done by the external agent from x_1 to x_2 is

$$W_{\text{ext}} = \int_{x_1}^{x_2} F_{\text{ext}}(x)\,dx = \int_{x_1}^{x_2} 2kx\,dx = \boxed{k(x_2^2 - x_1^2)}. \tag{131}$$

Since the external force is not balanced by the spring force, the kinetic energy of the mass can change. Assuming no other forces do work, the work-energy theorem gives

$$\Delta K = \int_{x_1}^{x_2} F_{\text{net}}(x)\,dx = \int_{x_1}^{x_2} \left[F_{\text{ext}}(x) + F_s(x)\right]\,dx = W_{\text{ext}} + W_s \tag{132}$$

where

$$W_s = \int_{x_1}^{x_2} (-kx)\,dx = -\frac{1}{2}k(x_2^2 - x_1^2). \tag{133}$$

Hence,

$$\boxed{\Delta K = \frac{1}{2}k(x_2^2 - x_1^2)}. \tag{134}$$

Thus, half of the external work changes the spring's potential energy, while the remaining half changes the kinetic energy of the mass:

$$W_{\text{ext}} = \Delta U + \Delta K. \tag{135}$$

5.2 Simple (mathematical) pendulum

A mass m is suspended from a fixed point by a light, inextensible string of length L as shown in Figure 14. The mass is displaced to an angle θ_0 from the vertical and released from rest.

(a) Write down the equation of motion for the pendulum.

(b) Determine the period of oscillation for small angles.

(c) How the angular displacement, angular velocity and angular acceleration of the pendulum vary with time for small angles?

Solution:

(a) Since the particle always moves along a circular arc, thinking in terms of the rotational form of Newton's second law is more convenient. The torque about the pivot point is given by

$$\tau = -mgL\sin\theta, \tag{136}$$

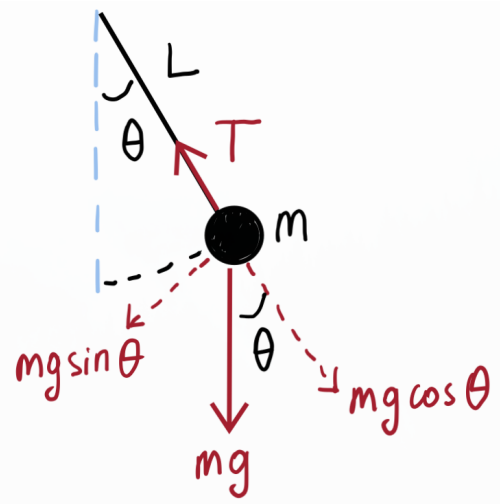


Figure 14: Simple pendulum. A mass m is suspended from a fixed point by a light, inextensible string of length L. The mass is displaced to an angle θ_0 from the vertical and released from rest.

where the negative sign indicates that the torque acts to restore the pendulum to its equilibrium position. The moment of inertia of the mass about the pivot point is

$$I = mL^2. \tag{137}$$

Therefore, the equation of motion for the pendulum is

$$I\ddot{\theta} = \tau \implies mL^2\ddot{\theta} = -mgL\sin\theta \implies \ddot{\theta} + \frac{g}{L}\sin\theta = 0. \tag{138}$$

(b) For small angles, we can use the small-angle approximation $\sin\theta \approx \theta$. Therefore, the equation of motion becomes

$$\ddot{\theta} + \frac{g}{L}\theta = 0. \tag{139}$$

This is a simple harmonic oscillator equation with angular frequency

$$\omega_0 = \sqrt{\frac{g}{L}}. \tag{140}$$

The period of oscillation is given by

$$T = \frac{2\pi}{\omega_0} = 2\pi\sqrt{\frac{L}{g}}. \tag{141}$$

(c) Since Eq. (139) is a simple oscillator equation, we already know the general solution:

$$\theta(t) = \theta_0\cos(\omega_0 t), \tag{142}$$

where θ_0 is the initial angular displacement. The angular velocity as a function of time is given by

$$\omega(t) = \dot{\theta}(t) = -\theta_0 \omega_0 \sin(\omega_0 t). \tag{143}$$

The angular acceleration as a function of time is given by

$$\alpha(t) = \dot{\omega}(t) = -\theta_0 \omega_0^2 \cos(\omega_0 t). \tag{144}$$

5.3 Phase Space Trajectory of a Damped Harmonic Pendulum

Write down the equation of motion for a damped harmonic pendulum without small angle approximation. Then, plot the phase space trajectory of the pendulum in the θ-$\dot{\theta}$ plane. Explain the behavior of the trajectory and how it relates to the damping of the system.

Solution: We know from Exercise 5.2 that the equation of motion for a simple pendulum is

$$\ddot{\theta} + \frac{g}{L} \sin\theta = 0. \tag{145}$$

If we introduce a damping term proportional to the angular velocity, the equation of motion becomes

$$\ddot{\theta} + 2\gamma\dot{\theta} + \frac{g}{L} \sin\theta = 0, \tag{146}$$

where γ is the damping constant. This is a second-order nonlinear differential equation that describes the motion of a damped harmonic pendulum without small angle approximation. If we represent the state of the pendulum with:

$$\mathbf{\Theta} = \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix}, \tag{147}$$

we can rewrite the equation of motion as a system of first-order differential equations:

$$\begin{bmatrix} \dot{\theta} \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} \dot{\theta} \\ -2\gamma\dot{\theta} - \frac{g}{L}\sin\theta \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{g}{L}\frac{\sin\theta}{\theta} & 0 \end{bmatrix} \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix}. \tag{148}$$

Note that the system is nonlinear due to the $\sin\theta$ term. In other words, in the equation

$$\dot{\mathbf{\Theta}} = \hat{S}\mathbf{\Theta}, \tag{149}$$

the operator $\hat{S}$ is a function of the state vector $\mathbf{\Theta}$. Therefore, the phase space trajectory of the pendulum will depend on the initial conditions and the damping constant γ. For small angles, the system can be approximated as linear, and the phase space trajectory will be an ellipse. However, for larger angles, the trajectory will deviate from an ellipse and become more complex.

The phase space trajectory of the pendulum can be plotted by numerically solving the system of equations for different initial conditions and plotting $\dot{\theta}$ versus θ. The resulting trajectory will show how the angular displacement and angular velocity evolve over time as shown in

Fig. 15. The trajectory will spiral inward toward the origin, indicating that the pendulum is losing energy due to damping and eventually coming to rest at the equilibrium position. The rate at which the trajectory spirals inward will depend on the damping constant γ. A larger damping constant will result in a faster decay of the oscillations and a quicker approach to equilibrium.

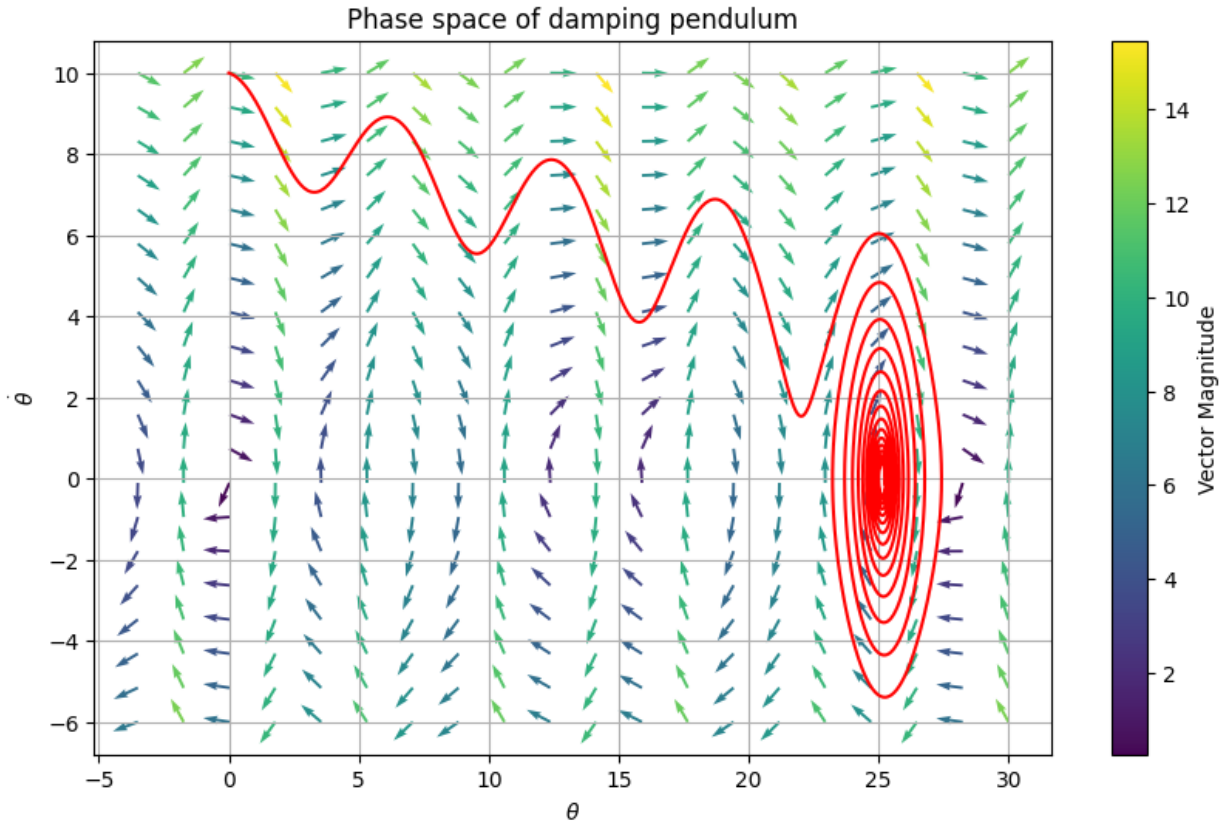


Figure 15: Phase trajectory of a damped harmonic pendulum in phase space with initial conditions $\theta_0 = 0$ and $\dot{\theta}_0 = 10$. The oscillating trajectory falls into a spiral pattern indicating that the pendulum makes 4 full rotations as it loses its energy due to damping and eventually coming to rest at the equilibrium position.

5.4 Crossing the Equilibrium Position in Critically Damped Motion

Consider a critically damped oscillator with damping constant $\gamma = 1\,\mathrm{s}^{-1}$. Initially, the particle is at $x_0 = 1\,\mathrm{m}$ and has velocity $v_0 = -2\,\mathrm{m/s}$.

(a) Determine the position of the particle as a function of time.

(b) Find the time at which the particle crosses the equilibrium position.

(c) Determine the maximum displacement of the particle on the opposite side of equilibrium and explain why the motion is not oscillatory.

Solution:

(a) The general solution for critically damped motion is

$$x(t) = [x_0 + (v_0 + \gamma x_0)t]\, e^{-\gamma t}. \tag{150}$$

Substituting the given initial conditions, we obtain

$$\boxed{x(t) = (1 - t)e^{-t},} \tag{151}$$

where t is measured in seconds and x in meters.

(b) The particle crosses the equilibrium position when $x(t) = 0$. Therefore,

$$(1 - t)e^{-t} = 0. \tag{152}$$

Since the exponential function never vanishes, we must have

$$1 - t = 0. \tag{153}$$

Thus, the particle crosses the equilibrium position at

$$\boxed{t = 1\,\text{s}.} \tag{154}$$

(c) To determine the maximum displacement on the opposite side of equilibrium, we find when the velocity vanishes. Differentiating the position with respect to time gives

$$v(t) = \dot{x}(t) = (t - 2)e^{-t}. \tag{155}$$

Setting $v(t) = 0$, we obtain

$$(t - 2)e^{-t} = 0, \tag{156}$$

which gives

$$t = 2\,\text{s}. \tag{157}$$

Substituting this into the position function, we find

$$x(2) = (1 - 2)e^{-2} = -e^{-2}\,\text{m}. \tag{158}$$

Therefore, the particle reaches its maximum displacement on the opposite side of equilibrium at

$$\boxed{x_{\min} = -e^{-2}\,\text{m} \approx -0.135\,\text{m}.} \tag{159}$$

The particle initially moves toward equilibrium, crosses it at $t = 1\,\text{s}$, and continues into the negative region. At $t = 2\,\text{s}$, it momentarily stops and reverses direction, approaching equilibrium from below as $t \to \infty$.

Although the particle crosses the equilibrium position once, the motion is not oscillatory because it never crosses equilibrium again. This illustrates that critical damping does not necessarily prevent a particle from crossing equilibrium; rather, it prevents repeated oscillations about equilibrium.

5.5 Kinetic energy resonance in a driven damped harmonic oscillator

At which driving frequency ω does the average kinetic energy of a driven damped harmonic oscillator reach its maximum?

Solution:

The steady-state displacement of a driven damped harmonic oscillator is

$$x(t) = A\cos(\omega t + \phi), \tag{160}$$

where ω is the driving frequency, A is the displacement amplitude, and ϕ is the phase. The corresponding velocity is

$$v(t) = \dot{x}(t) = -\omega A\sin(\omega t + \phi). \tag{161}$$

Therefore, the instantaneous kinetic energy is

$$T(t) = \frac{1}{2}mv^2(t) = \frac{1}{2}m\omega^2 A^2\sin^2(\omega t + \phi). \tag{162}$$

The average kinetic energy over one complete period $P = 2\pi/\omega$ is defined as

$$\langle T\rangle = \frac{1}{P}\int_0^P T(t)\,dt. \tag{163}$$

Substituting the instantaneous kinetic energy gives

$$\langle T\rangle = \frac{m\omega^2 A^2}{2P}\int_0^P \sin^2(\omega t + \phi)\,dt \tag{164}$$

$$= \frac{m\omega^2 A^2}{4P}\int_0^P [1 - \cos(2\omega t + 2\phi)]\,dt \tag{165}$$

$$= \frac{1}{4}m\omega^2 A^2, \tag{166}$$

where we used the identity $\sin^2\theta = (1 - \cos 2\theta)/2$ and the fact that the cosine term averages to zero over a complete period. The steady-state displacement amplitude is

$$A = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\gamma^2\omega^2}}, \tag{167}$$

where F_0 is the amplitude of the driving force, ω_0 is the natural frequency, and $\gamma > 0$ is the damping constant. Substituting this expression into the average kinetic energy, we obtain

$$\langle T\rangle = \frac{F_0^2}{4m}\frac{\omega^2}{(\omega_0^2 - \omega^2)^2 + 4\gamma^2\omega^2}. \tag{168}$$

For $\omega > 0$, dividing the numerator and denominator by ω^2 gives

$$\langle T\rangle = \frac{F_0^2}{4m}\frac{1}{(\omega_0^2/\omega - \omega)^2 + 4\gamma^2}. \tag{169}$$

The average kinetic energy is maximized when the denominator is minimized. Since the squared term is nonnegative, its minimum occurs when

$$\frac{\omega_0^2}{\omega} - \omega = 0. \tag{170}$$

Thus, the driving frequency at which the average kinetic energy reaches its maximum is

$$\boxed{\omega_{\text{res}} = \omega_0.} \tag{171}$$

Conclusion: The kinetic energy resonance occurs at the natural frequency ω_0, independently of the damping constant γ. This differs from displacement resonance, which occurs at

$$\omega_{\text{res}}^{(x)} = \sqrt{\omega_0^2 - 2\gamma^2}, \tag{172}$$

provided $\gamma < \omega_0/\sqrt{2}$. This means that the energy pumped into the system is maximized when the driving frequency is set to the natural frequency ω_0. That's the reason why we call the natural frequency ω_0 the resonance frequency of the system; in resonance, the system absorbs energy most efficiently from the driving force.

5.6 Amplitude damping

The amplitude of a lightly damped oscillator decreases by 5% each cycle. What percentage of mechanical energy is lost per cycle?

Solution: The mechanical energy of a harmonic oscillator is proportional to the square of its amplitude. If the amplitude decreases by 5% each cycle, the new amplitude after one cycle is

$$A_{\text{new}} = 0.95 A_{\text{old}}. \tag{173}$$

The mechanical energy after one cycle is

$$E_{\text{new}} \propto A_{\text{new}}^2 = (0.95 A_{\text{old}})^2 = 0.9025 A_{\text{old}}^2. \tag{174}$$

Therefore, the percentage of mechanical energy lost per cycle is

$$\text{Percentage lost} = \left(1 - \frac{E_{\text{new}}}{E_{\text{old}}}\right) \times 100\% = (1 - 0.9025) \times 100\% = 9.75\%. \tag{175}$$

5.7 Coupled Oscillators

Consider two masses m_1 and m_2 connected by a spring with spring constant k, moving along a frictionless horizontal surface. Let x_1 and x_2 denote their displacements from their respective equilibrium positions. Show that the relative displacement $x = x_2 - x_1$ undergoes simple harmonic motion, and determine its angular frequency.

Solution:

We choose the positive x-direction to be to the right. The key observation is that the force exerted by the spring depends only on the relative displacement of the two masses, not on their individual positions. Suppose that mass m_1 moves to the right by x_1, while mass m_2 moves to the right by x_2. The change in the length of the spring is then

$$\Delta L = x_2 - x_1.$$

Indeed, if $x_2 > x_1$, the separation between the masses increases and the spring is stretched. If $x_2 < x_1$, the separation decreases and the spring is compressed. If $x_2 = x_1$, both masses have moved by the same amount, so the length of the spring remains unchanged.

To determine the forces, first suppose that the spring is stretched, so that $x_2 - x_1 > 0$. Since the spring tends to return to its natural length, it pulls m_1 to the right and m_2 to the left. According to Hooke's law, the forces acting on the two masses are therefore

$$F_1 = k(x_2 - x_1), \qquad F_2 = -k(x_2 - x_1).$$

These expressions also apply when the spring is compressed, since $x_2 - x_1 < 0$ automatically reverses the directions of both forces. Note that $F_1 = -F_2$, in agreement with Newton's third law. Applying Newton's second law to each mass, we obtain the coupled equations of motion:

$$\begin{aligned} m_1\ddot{x}_1 &= k(x_2 - x_1), \\ m_2\ddot{x}_2 &= -k(x_2 - x_1). \end{aligned}$$

We now introduce the relative displacement

$$x = x_2 - x_1.$$

Differentiating twice with respect to time gives

$$\ddot{x} = \ddot{x}_2 - \ddot{x}_1.$$

Using the equations of motion, we find

$$\ddot{x}_1 = \frac{k}{m_1}x, \qquad \ddot{x}_2 = -\frac{k}{m_2}x.$$

Substituting these expressions into the equation for the relative acceleration yields

$$\ddot{x} = -\frac{k}{m_2}x - \frac{k}{m_1}x,$$

or

$$\boxed{\ddot{x} + k\left(\frac{1}{m_1} + \frac{1}{m_2}\right)x = 0.}$$

This has the standard form of the simple harmonic oscillator equation,

$$\ddot{x} + \omega^2 x = 0.$$

Therefore, the relative displacement undergoes simple harmonic motion with angular frequency

$$\boxed{\omega = \sqrt{k\left(\frac{1}{m_1} + \frac{1}{m_2}\right)}.}$$

Physical interpretation:

The spring does not respond to the absolute positions of the masses, but only to their separation. When the spring is stretched, it pulls the masses toward each other; when compressed, it pushes them apart. In either case, the spring produces a restoring force that tends to bring the relative displacement back to zero. Since this force is proportional to the relative displacement, the resulting relative motion is simple harmonic.

Notice also that if both masses move by the same amount, $x_1 = x_2$, the spring is neither stretched nor compressed, and hence exerts no force. Thus, the two masses can move together with constant velocity without oscillating. More generally, the motion of the system can be separated into two independent parts: uniform motion of the center of mass and oscillatory motion of the masses relative to each other.

5.8 Determining the damping constant from amplitude decay

The amplitude of a damped harmonic oscillator reduces to the half of its initial value in 1 second. Find the damping constant γ.

Solution: The amplitude of a damped harmonic oscillator is given by

$$A(t) = A_0 e^{-\gamma t}. \tag{176}$$

If the amplitude reduces to half its initial value in 1 second, then

$$\frac{1}{2}A_0 = A_0 e^{-\gamma \cdot 1}. \tag{177}$$

Simplifying,

$$\frac{1}{2} = e^{-\gamma}. \tag{178}$$

Taking the natural logarithm of both sides,

$$\ln\left(\frac{1}{2}\right) = -\gamma. \tag{179}$$

Therefore,

$$\boxed{\gamma = \ln(2)}. \tag{180}$$

5.9 Spring Compression with and without Friction

A block of mass m slides on a horizontal surface toward an ideal massless spring of spring constant k attached to a wall. At $t = 0$, the block first touches the initially uncompressed spring with speed v_0. Determine the maximum compression of the spring:

(a) if the floor is frictionless;

(b) if the coefficient of kinetic friction between the block and the floor is μ_k.

Solution:

At maximum compression, the block is momentarily at rest.

(a) On a frictionless floor, mechanical energy is conserved. The initial kinetic energy is converted entirely into spring potential energy:

$$\frac{1}{2}mv_0^2 = \frac{1}{2}kx_{\max}^2. \tag{181}$$

Therefore,

$$\boxed{x_{\max} = v_0\sqrt{\frac{m}{k}}}. \tag{182}$$

(b) During compression, the kinetic friction force has constant magnitude $\mu_k mg$ and acts opposite to the block's motion. Its work is

$$W_{\text{friction}} = -\mu_k mgx_{\max}. \tag{183}$$

The work done by friction equals the change in mechanical energy:

$$-\mu_k mgx_{\max} = \frac{1}{2}kx_{\max}^2 - \frac{1}{2}mv_0^2. \tag{184}$$

Equivalently,

$$\frac{1}{2}mv_0^2 = \frac{1}{2}kx_{\max}^2 + \mu_k mgx_{\max}. \tag{185}$$

Rearranging gives the quadratic equation

$$kx_{\max}^2 + 2\mu_k mgx_{\max} - mv_0^2 = 0. \tag{186}$$

Taking the nonnegative root,

$$\boxed{x_{\max} = \frac{\sqrt{(\mu_k mg)^2 + kmv_0^2} - \mu_k mg}{k}}. \tag{187}$$

For $\mu_k = 0$, this expression reduces to the frictionless result.

5.10 Beats and Resonance in an Undamped Driven Oscillator

An undamped spring–mass system has mass m, spring constant k, and natural angular frequency

$$\omega_0 = \sqrt{\frac{k}{m}}. \tag{188}$$

Initially, the mass is at rest at equilibrium:

$$x(0) = 0, \qquad \dot{x}(0) = 0. \tag{189}$$

At $t = 0$, an external force is switched on:

$$F(t) = F_0 \cos(\omega t), \qquad t \geq 0, \tag{190}$$

where $F_0 > 0$ and $\omega > 0$.

(a) Find $x(t)$ for $\omega \neq \omega_0$, satisfying both initial conditions.

(b) For $|\omega - \omega_0| \ll \omega_0$, rewrite the solution to identify the rapid oscillation and the slowly varying amplitude envelope. Find the time between successive maxima of this envelope.

(c) Take the limit $\omega \to \omega_0$ to obtain the motion at exact resonance. Describe how the amplitude changes with time.

Hint: You may use

$$\cos a - \cos b = -2 \sin\left(\frac{a+b}{2}\right) \sin\left(\frac{a-b}{2}\right). \tag{191}$$

Also, the general amplitude expression for driven oscillations is given by

$$A(\omega) = \frac{F_0}{m\sqrt{(\omega_0^2 - \omega^2)^2 + (2\gamma\omega)^2}}, \tag{192}$$

Solution.

(a) The equation of motion is

$$\ddot{x} + \omega_0^2 x = \frac{F_0}{m} \cos(\omega t). \tag{193}$$

For $\omega \neq \omega_0$, a particular solution is

$$x_{\mathrm{p}}(t) = \frac{F_0}{m(\omega_0^2 - \omega^2)} \cos(\omega t). \tag{194}$$

Adding the homogeneous solution gives

$$x(t) = C \cos(\omega_0 t) + D \sin(\omega_0 t) + \frac{F_0}{m(\omega_0^2 - \omega^2)} \cos(\omega t). \tag{195}$$

The initial conditions require

$$C = -\frac{F_0}{m(\omega_0^2 - \omega^2)}, \qquad D = 0. \tag{196}$$

Hence,

$$\boxed{x(t) = \frac{F_0}{m(\omega_0^2 - \omega^2)} \left[\cos(\omega t) - \cos(\omega_0 t)\right].} \tag{197}$$

(b) Applying the trigonometric identity gives

$$x(t) = \frac{2F_0}{m(\omega^2 - \omega_0^2)} \sin\left(\frac{\omega + \omega_0}{2} t\right) \sin\left(\frac{\omega - \omega_0}{2} t\right). \tag{198}$$

When $|\omega - \omega_0| \ll \omega_0$, the first sine describes a rapid oscillation with angular frequency

$$\omega_{\text{rapid}} = \frac{\omega + \omega_0}{2} \approx \omega_0. \tag{199}$$

The slowly varying, nonnegative amplitude envelope is

$$A_{\text{env}}(t) = \frac{2F_0}{m|\omega^2 - \omega_0^2|} \left|\sin\left(\frac{\omega - \omega_0}{2} t\right)\right|. \tag{200}$$

Successive maxima of $|\sin u|$ are separated by π in its argument. Therefore, the beat period is

$$\boxed{T_{\text{beat}} = \frac{2\pi}{|\omega - \omega_0|}.} \tag{201}$$

(c) Taking the limit at fixed t, using l'Hopital's rule, gives

$$\lim_{\omega \to \omega_0} \frac{\cos(\omega t) - \cos(\omega_0 t)}{\omega_0^2 - \omega^2} = \lim_{\omega \to \omega_0} \frac{-t \sin(\omega t)}{-2\omega} = \frac{t \sin(\omega_0 t)}{2\omega_0}. \tag{202}$$

Thus, at exact resonance,

$$\boxed{x(t) = \frac{F_0}{2m\omega_0} t \sin(\omega_0 t).} \tag{203}$$

The amplitude envelope grows linearly:

$$\boxed{A_{\text{env}}(t) = \frac{F_0}{2m\omega_0} t.} \tag{204}$$

There is no bounded steady-state response at exact resonance in this ideal undamped model. The external force continually supplies energy to the oscillator.

5.11 Simple pendulum with a horizontally oscillating pivot

A simple pendulum consists of a point mass m suspended from a massless, inextensible string of length L. Its pivot undergoes prescribed horizontal motion

$$d(t) = d_0 \cos(\omega t),$$

where d_0 is the displacement amplitude and ω is the driving angular frequency. Neglect air resistance and friction at the pivot, and assume that the string remains taut. Let $\theta(t)$ denote the angular displacement from the downward vertical, positive toward the positive horizontal direction.

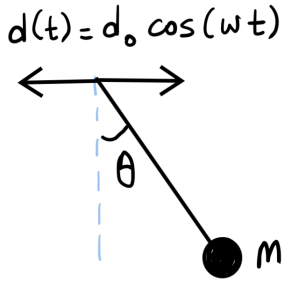


Figure 16: A pendulum of mass m and length L is driven by a pivot that oscillates horizontally with displacement $d(t) = d_0 \cos(\omega t)$.

Derive the equation of motion in the small-angle approximation and find the periodic forced response for $\omega \neq \sqrt{g/L}$. Explain the meaning of "steady state" in the absence of damping and discuss what happens at resonance.

Solution. Choose an inertial coordinate system with x horizontal and y vertically upward. Taking the pivot height to be $y = 0$, the position of the pendulum bob is

$$x(t) = d(t) + L \sin\theta(t), \qquad y(t) = -L \cos\theta(t).$$

Differentiating once gives

$$\dot{x} = \dot{d} + L\dot{\theta} \cos\theta, \qquad \dot{y} = L\dot{\theta} \sin\theta.$$

Differentiating again gives

$$\ddot{x} = \ddot{d} + L\ddot{\theta} \cos\theta - L\dot{\theta}^2 \sin\theta,$$
$$\ddot{y} = L\ddot{\theta} \sin\theta + L\dot{\theta}^2 \cos\theta.$$

The unit vector tangent to the circular path relative to the pivot, pointing in the direction of increasing θ, is

$$\mathbf{e}_\theta = (\cos\theta, \sin\theta).$$

The tangential component of the bob's acceleration is therefore

$$\begin{aligned} a_\theta &= \ddot{x}\cos\theta + \ddot{y}\sin\theta \\ &= \ddot{d}\cos\theta + L\ddot{\theta}\cos^2\theta - L\dot{\theta}^2\sin\theta\cos\theta \\ &\quad + L\ddot{\theta}\sin^2\theta + L\dot{\theta}^2\cos\theta\sin\theta \\ &= \ddot{d}\cos\theta + L\ddot{\theta}. \end{aligned}$$

Here the terms proportional to $\dot{\theta}^2$ cancel because the corresponding acceleration is radial. The string tension is radial and has no tangential component. The tangential component of gravity is

$$F_\theta = (0, -mg)\cdot\mathbf{e}_\theta = -mg\sin\theta.$$

Consequently, Newton's second law gives

$$m\left(L\ddot{\theta} + \ddot{d}\cos\theta\right) = -mg\sin\theta,$$

or

$$\ddot{\theta} + \frac{g}{L}\sin\theta = -\frac{\ddot{d}}{L}\cos\theta.$$

Since

$$\dot{d} = -d_0\omega\sin(\omega t), \qquad \ddot{d} = -d_0\omega^2\cos(\omega t),$$

the exact equation of motion is

$$\boxed{\ddot{\theta} + \frac{g}{L}\sin\theta = \frac{d_0\omega^2}{L}\cos(\omega t)\cos\theta.}$$

Small-angle approximation. For $|\theta| \ll 1$, with θ measured in radians,

$$\sin\theta = \theta - \frac{\theta^3}{6} + \cdots, \qquad \cos\theta = 1 - \frac{\theta^2}{2} + \cdots.$$

Retaining only terms through first order in θ gives

$$\boxed{\ddot{\theta} + \omega_0^2\theta = \frac{d_0\omega^2}{L}\cos(\omega t), \qquad \omega_0 = \sqrt{\frac{g}{L}}.}$$

The neglected restoring term starts at order θ^3, while the neglected correction to the driving term starts at order θ^2. No approximation has been made to the prescribed pivot motion.

Periodic forced response away from resonance. For $\omega \neq \omega_0$, seek a particular solution of the form

$$\theta_{\mathrm{p}}(t) = A\cos(\omega t).$$

Its second derivative is

$$\ddot{\theta}_{\mathrm{p}}(t) = -A\omega^2\cos(\omega t).$$

Substituting into the linearized equation gives

$$-A\omega^2\cos(\omega t) + \omega_0^2 A\cos(\omega t) = \frac{d_0\omega^2}{L}\cos(\omega t).$$

Matching the coefficients of $\cos(\omega t)$ yields

$$A(\omega_0^2 - \omega^2) = \frac{d_0\omega^2}{L}.$$

Thus the periodic forced response is

$$\boxed{\theta_{\mathrm{p}}(t) = \frac{d_0\omega^2}{L(\omega_0^2 - \omega^2)}\cos(\omega t) = \frac{d_0\omega^2}{g - L\omega^2}\cos(\omega t).}$$

Its angular amplitude is

$$\Theta = \frac{d_0\omega^2}{|g - L\omega^2|}.$$

For $\omega < \omega_0$, the angular displacement is in phase with the pivot displacement. For $\omega > \omega_0$, it is opposite in phase.

The homogeneous equation,

$$\ddot{\theta}_{\mathrm{h}} + \omega_0^2\theta_{\mathrm{h}} = 0,$$

has the solution

$$\theta_{\mathrm{h}}(t) = C\cos(\omega_0 t) + D\sin(\omega_0 t).$$

Hence the general solution is

$$\boxed{\theta(t) = C\cos(\omega_0 t) + D\sin(\omega_0 t) + \frac{d_0\omega^2}{g - L\omega^2}\cos(\omega t).}$$

The constants C and D are determined by the initial conditions. For example, if $\theta(0) = \theta_i$ and $\dot{\theta}(0) = v_i$, then

$$C = \theta_i - \frac{d_0\omega^2}{g - L\omega^2}, \qquad D = \frac{v_i}{\omega_0}.$$

Without damping, the homogeneous oscillation persists indefinitely. Therefore, the periodic forced response is not a uniquely selected late-time steady state. It is the solution obtained when the free oscillation is absent. In a weakly damped system, the free oscillation decays, and the late-time response approaches the expression above as damping tends to zero, provided the driving frequency remains away from resonance.

The small-angle approximation requires the total angular displacement to remain small. For the periodic forced response alone, this requires

$$\boxed{\frac{d_0\omega^2}{|g - L\omega^2|} \ll 1.}$$

If free oscillations are present, their amplitude must also be small.

Resonance. At $\omega = \omega_0$, the linearized equation becomes

$$\ddot{\theta} + \omega_0^2 \theta = \frac{d_0 \omega_0^2}{L} \cos(\omega_0 t).$$

The previous periodic ansatz fails because it solves the homogeneous equation. Instead, try

$$\theta_{\mathrm{p}}(t) = Bt \sin(\omega_0 t).$$

Differentiating gives

$$\dot{\theta}_{\mathrm{p}} = B \sin(\omega_0 t) + B\omega_0 t \cos(\omega_0 t),$$
$$\ddot{\theta}_{\mathrm{p}} = 2B\omega_0 \cos(\omega_0 t) - B\omega_0^2 t \sin(\omega_0 t).$$

Therefore,

$$\ddot{\theta}_{\mathrm{p}} + \omega_0^2 \theta_{\mathrm{p}} = 2B\omega_0 \cos(\omega_0 t).$$

Matching coefficients gives

$$2B\omega_0 = \frac{d_0 \omega_0^2}{L}, \qquad B = \frac{d_0 \omega_0}{2L}.$$

The resonant particular solution is consequently

$$\boxed{\theta_{\mathrm{p}}(t) = \frac{d_0 \omega_0}{2L}\, t \sin(\omega_0 t).}$$

There is no bounded periodic forced response at resonance in the undamped linear model. The growing amplitude eventually violates the small-angle approximation.